

Difficulty in artificial word learning impacts targeted memory reactivation and its underlying neural signatures

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Abstract

Sleep associated memory consolidation and reactivation play an important role in language acquisition and learning of new words. However, it is unclear to what extent properties of word learning difficulty impact sleep associated memory reactivation. To address this gap, we investigated in twenty-two young healthy adults the effectiveness of auditory targeted memory reactivation (TMR) during non-rapid eye movement sleep of artificial words with easy and difficult to learn phonotactical properties. Here, we found that TMR of the easy words improved their overnight memory performance, whereas TMR of the difficult words had no effect. By comparing EEG activities after TMR presentations, we found an increase in slow wave density independent of word difficulty, whereas the spindle-band power nested during the slow wave up-states - as an assumed underlying activity of memory reactivation - was significantly higher in the easy/effective compared to the difficult/ineffective condition. Our findings indicate that word learning difficulty by phonotactics impacts the effectiveness of TMR and further emphasize the critical role of prior encoding depth in sleep associated memory reactivation.

eLife assessment

This study provides **useful** findings on how phonetic properties of words, i.e., their difficulty and prior knowledge, influence the outcome of targeted memory reactivation (TMR) during sleep. While these findings are supported by **solid** evidence, they are based on a small sample size warranting future work to shed further light on the impact of TMR in language learning.

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Introduction

Learning of new words is essential in language acquisition. During the learning process of unknown words, top-down influences can facilitate perception and encoding of these words depending on the similarity between the new words and the pre-existing knowledge. One critical

characteristic of similarity to pre-existing knowledge in auditory word processing is its speech sound (phoneme) pattern. In phonology as the field of language specific phoneme structures, phonotactics determines the constraints of word phoneme composition of a specific language. Previous studies have shown that artificial words with similar phonotactic properties to real words are easier to learn and to remember^{1,2}, suggesting that pre-existing knowledge about the sound structure (i.e. schemas) facilitates encoding and consolidation of new words. A synchronized interplay between the ventromedial prefrontal cortex, hippocampus and unimodal associative cortices is proposed to explain schema instantiation and schema mnemonic effects³ at the neural level. To describe the influence of phonotactics on memory processing within this framework, new words with phoneme structures congruent to activated schema variables are inserted with less cognitive effort and difficulty into pre-existing knowledge networks of long-term memory in contrast to schema-incongruent words.

According to the complementary system account by Gaskell and colleagues^{4,5}, learning of new words comprises a first stage of rapid initial familiarization represented by neural activities of the medial temporal lobe and a second stage of slow offline consolidation of the neocortex. The model suggests that, after initial hippocampal learning, through sleep consolidation memory representations, the words are gradually integrated into long-term lexical and phonological memory structures of the neocortex. Additionally, these processes of memory consolidation might depend on spontaneous and repeated reactivation of the newly formed memory traces during non-rapid eye-movement (NREM) sleep⁶. The active system consolidation hypothesis proposes synchronized hippocampal sharp-wave-ripples, thalamocortical sleep spindles and slow oscillations of NREM sleep as underlying activities of memory reactivation. Accordingly, studies in humans⁷⁻⁹ and in rodent models^{10,11} found that sleep spindles and hippocampal sharp wave ripples occur grouped during slow oscillations and that these events correlate with memory consolidation^{6,12-15}.

While similarity of the learning material to pre-existing knowledge clearly facilitates encoding, this notion is controversial for the effects on sleep consolidation. For example, Havas and co-authors¹⁶ tested learning and consolidation of artificial words phonologically derived either from participant's native language (L1 words in Spanish) or foreign language (L2 words in Hungarian). Here, sleep-associated improvements in memory recognition were only observable in the condition of the L2 words, suggesting that a reduced similarity to pre-existing knowledge facilitates consolidation during sleep. Also, in a study by Payne et al.¹⁷, sleep-associated memory benefits occurred only for semantically unrelated word-pairs, but not for semantically related words.

On the other hand, Zion and colleagues¹⁸ reported that sleep facilitated consolidation in a second language learning task in participants with a higher degree of meta-linguistic knowledge, indicating a beneficial effect of pre-existing knowledge on new word consolidation. Similarly, Durrant and colleagues¹⁹ reported that only schema-conformant but not non-conformant learnings benefitted from sleep. In a study from our own lab, German-participants learning unknown Dutch vocabulary profited more from sleep compared to native French speaker when given the same amount of learning trials, whereas German native speakers might have more pre-existing knowledge about the Dutch words in relation to the French native speakers, which facilitated learning and consolidation of the new Dutch vocabulary²⁰.

A promising approach to examine sleep-associated memory consolidation is to experimentally bias reactivation by re-presenting olfactory²¹⁻²³ or auditory^{24,25} reminder cues during sleep, a technique known as targeted memory reactivation (TMR). TMR has been related to elevated slow-wave and spindle activity following cueing presentations as neural markers of memory reactivation^{26,27}. So far two studies suggest that prior knowledge increase the effectiveness of TMR: Groch and colleagues²⁸ reported improved memory performance by TMR for associations between pseudo-words and familiar objects, but not for word associations of

new/unknown objects. Here, before sleep, subjects showed as well better encoding performance for the familiar compared to the unknown objects. In another study²⁹ of learning sound-object locations, the effectiveness of TMR correlated positively with the initial encoding performance prior to sleep. Additionally, both studies^{28,29} found significant associations between TMR's effectiveness and sleep spindle activities after the cueing presentations. These findings suggest that higher similarity between the learning material and the pre-existing knowledge facilitates reactivation and consolidation during sleep. However, it is unknown whether these findings can be generalized to learning of new artificial words with different levels of phonotactical similarity to prior word-sound knowledge, as a factor of learning difficulty.

To address this issue, we designed artificial words with varying difficulty levels by phonotactical properties. To consider the beneficial effect of reward related information on sleep dependent memory consolidation and reactivation^{30–33}, we trained healthy young participants to categorize these words into rewarded and unrewarded words to gain and to avoid losses of money points. By using auditory TMR, we re-presented the artificial words during subsequent sleep to study its effect on categorization performance. As a manipulation check, we expect that artificial words with more familiar phonotactical properties would be easier to learn compared to words with less familiarity. Regarding previous findings^{28,29}, we hypothesized higher effectiveness of TMR for the easy to learn words in comparison to the difficult words, accompanied by higher oscillatory activity on the slow-wave and spindle range.

Results

Experimental design

To study the impact of difficulty in word learning on TMR, we developed a novel learning paradigm. We formed four sets of artificial words (40 words per set; see **Table S3** and **S4**) consisting of different sequences of two vowels and two consonants. Here, we subdivided the alphabet into two groups of consonants (C1: b, c, d, f, g, h, j, k, l, m; C2: n, p, q, r, s, t, v, w, x, z) and vowels (V1: a, e, i; V2: o, u, y). Four-letter-words were created by selecting letters from the vowel and consonant groups according to four different sequences (G1: C1, V1, V2, C2; G2: C1, V1, C2, V2; G3: V1, C1, C2, V2; G4: V1, C1, V2, C2; **Fig. 1a**; see methods for further details). Comparison analyses between the sets revealed significant differences in phonotactic probability (PP; **Fig. 1b**; unpaired *t*-tests: $G1 / G2 > G3 / G4$, $p < 0.005$, values of Cohen's $d > 0.71$). PP quantifies the frequency of a single phoneme (phoneme probabilities) or a sequence of phonemes (e.g., biphone probabilities) within a language and thus serves as a measurement of the similarity between the artificial words and the pre-existing real word knowledge. According to distinct levels of PP, we paired the four sets to the high- (G1 & G2) and low-PP (G3 & G4) condition, respectively.

During encoding, we trained the subjects to discriminate the artificial words based on reward associations by manual button presses. Therefore, the high- and the low-PP condition, consisted of 40 rewarded and 40 unrewarded words. As a two alternative forced-choice task, we assigned left- and right-hand button presses to the rewarded and the unrewarded word category, counterbalanced across subjects. We instructed the participants to respond to each word by left- or right-hand button presses, whereas one button means the word is rewarded (gain of money points) and the other button means the word is unrewarded (avoid the loss of money points). In the beginning, they had to guess. By three presentations of each word in randomized order and by feedback at the end of each trial, they learned to respond correctly according to the rewarded/unrewarded associations of the words (**Fig. 1c**).

To obtain a measurement of discrimination memory with respect to the potential influence of the response bias, we applied the signal detection theory³⁴. Because, we instructed the participants to respond to each word by left- or right-hand button presses and that one button means reward is

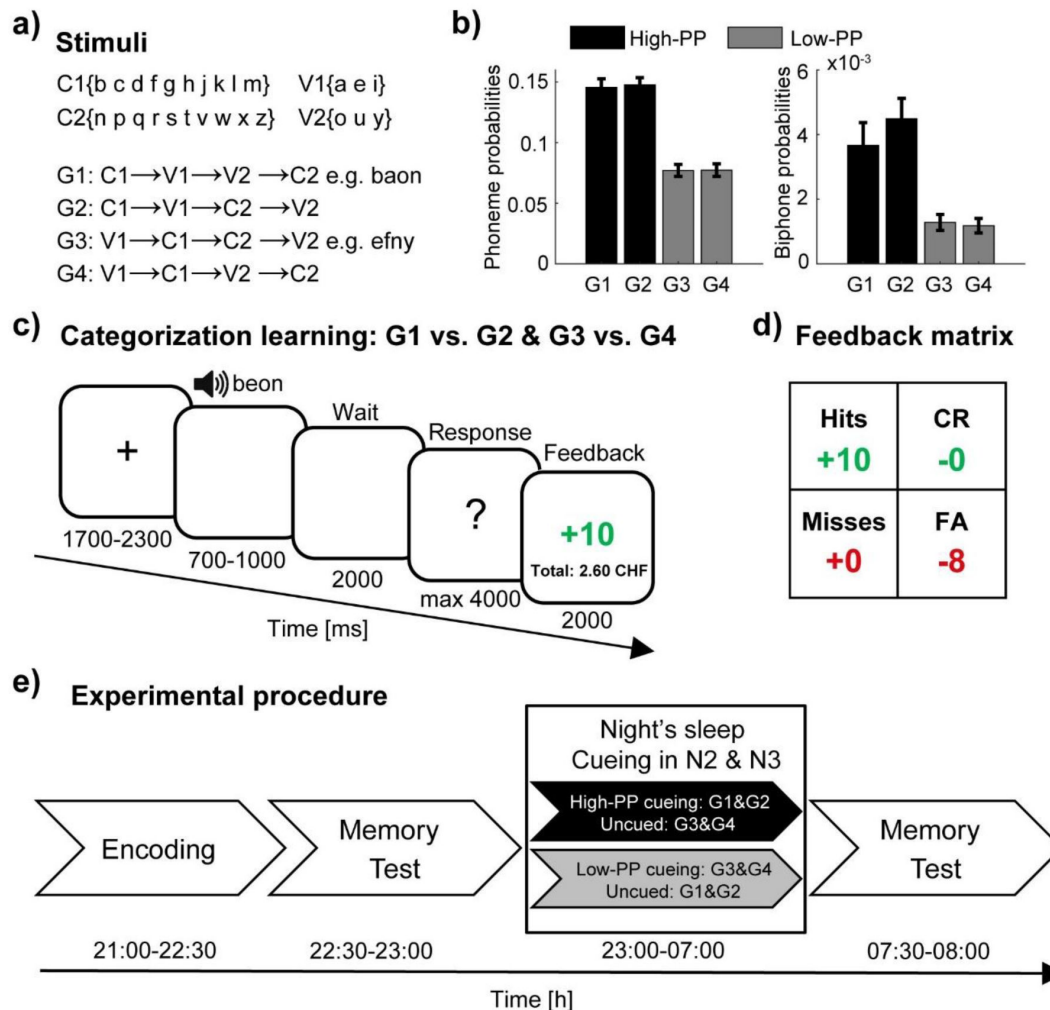


Fig. 1.

Experimental design.

a) Artificial words consist of two vowels and two consonants according to four different sequences. b) The phonotactic probabilities (PP) for single phonemes (left panel) and biphone probabilities (right panel) averaged values of the four sets (40 words per set) and pairing of the sets with respect to two distinct levels of PP in high- (black) and low-PP (gray). c) Schematic trial structure of the learning task with screen images and the durations in milliseconds. As a two alternative forced-choice task, responses of left- and right-hand button presses were assigned to the rewarded and the unrewarded word category, respectively. The participants were instructed to respond to each word by left- or right-hand button presses, whereas one button means the word is rewarded (gain of money points) and the other button means the word is unrewarded (avoid the loss of money points). d) Feedback matrix with the four answer types (hits: rewarded and correct; CR, correct rejections: unrewarded and correct; misses: rewarded and incorrect; FA, false alarms: unrewarded and incorrect) regarding to response and reward assignment of the word. Note, subjects could receive and lose money points dependent on correct and incorrect responses. e) Experimental procedure with experimental tasks and phases in temporal order. TMR took place in the NREM sleep stages 2 & 3. Error bars reflect standard errors of the mean (SEM).

present whereas the other button means reward is absent, we considered correct responses of words of the rewarded category as hits and words of the unrewarded category as correct rejections. Accordingly, we assigned the responses with regard to the reward associations of the words to the following four response types: hits (rewarded, correct); correct rejections (unrewarded, correct); misses (rewarded, incorrect); and false alarms (unrewarded, incorrect). Dependent on responses, subjects received money points (**Fig. 1d** [↗](#)).

After encoding and before sleep, we tested the pre-sleep memory performance (**Fig. 1e** [↗](#)). Subjects slept one night in the sleep laboratory and during the NREM sleep stages 2 & 3 we conducted auditory TMR of the low-PP words in one group of subjects and TMR of the high-PP words in the other group (between subject design). At the following morning, we tested the post-sleep memory performance.

Manipulation check based on encoding analyses

To validate our novel paradigm, we examined the influence of PP on encoding performance. Based on the signal detection theory³⁴ [↗](#), we calculated d' -values to measure subject's abilities to differentiate and categorize rewarded and unrewarded words of the two PP conditions over the three presentations of the encoding task (**Fig. 2a** [↗](#)). A repeated-measure ANOVA on d' with PP (high vs. low) and presentations (1 to 3) as within-subjects factors revealed significant main effects of PP ($F(1,32) = 5.13$, $p = 0.03$, $\eta^2 = 0.14$), and presentations ($F(2,64) = 95.67$, $p < 0.001$, $\eta^2 = 0.75$). Additional pairwise comparisons between PP conditions showed significant differences after the first presentation (paired t -tests of the three presentations in order: (1) $t(32) = 0.73$, $p = 0.47$; (2) $t(32) = 2.35$, $p = 0.03$, Cohen's $d = 0.41$; (3) $t(32) = 2.17$, $p = 0.04$, Cohen's $d = 0.38$). These results indicate superior learning performance for words with high- compared to words with low-PP.

To examine the oscillatory correlates of encoding performance, we conducted time-frequency analyses time-locked to auditory word representations (**Fig. 2b** [↗](#)) and extracted power values over time, frequency, and EEG electrodes by using wavelet transformations (see methods). This analysis revealed a strong desynchronization after word presentations (0.7 to 1.9s to stimulus onset) in the frequency range of the alpha waves (8-13 Hz) and pronounced over parietal and occipital electrodes in contrast to baseline activity (−1 to −0.1s to stimulus onset; **Fig. 2c** [↗](#)). Testing whether changes in alpha desynchronization correspond to increased encoding performance over repetitive presentations, a repeated-measure ANOVA on alpha power values merged over central electrodes (Pz, Cz) including presentations (1 to 3) as a single within-subjects factor revealed a significant increase in alpha desynchronization ($F(2,64) = 4.19$, $p = 0.02$, $\eta^2 = 0.12$).

Regarding distinct levels of encoding performance between high- and low-PP, we compared next changes in alpha desynchronization between these conditions. We obtained the changes in power values by subtracting the first from the second and third presentation for the high- and low-PP condition, respectively. Here, the first word presentation of naive stimulus processing served us with a more representative baseline condition covering the time-window of interest of 0.7 to 1.9 s after the stimulus onset to examine relevant changes of encoding. Cluster-based statistics of the third presentation revealed a significant cluster over the left posterior electrodes with a stronger alpha power decrease for the high- in contrast to the low-PP condition (time window: 0.7 to 1.9s; frequency range 8-13Hz; averaged t -value over the 5 cluster electrodes = -2.69 , $p = 0.01$, Cohen's $d = -0.48$; **Fig. 2d** [↗](#)). A post-hoc repeated-measure ANOVA on alpha power changes (merged over Pz and Cz electrodes) with PP (high vs. low) and presentations (2 to 3) as within-subjects factors revealed a main effect of PP ($F(1,32) = 5.42$, $p = 0.03$, $\eta^2 = 0.15$), and a significant interaction ($F(1,32) = 7.38$, $p = 0.01$, $\eta^2 = 0.19$; **Fig. 2e** [↗](#)). Additional post-hoc pairwise comparison between PP conditions showed significant difference of the third presentation (paired t -tests of presentations in order: (2) $t(32) = -0.36$, $p = 0.72$; (3) $t(32) = -3.55$, $p = 0.001$, Cohen's $d = -0.62$) and a t -test against 0 revealed significantly decreased activity in high-PP of the third presentation ($t(32) = -2.92$, $p = 0.006$, Cohen's $d = -0.51$). In addition to the behavior results, these EEG results indicate differences between PP conditions in desynchronization of alpha oscillations, as an assumed neural correlate

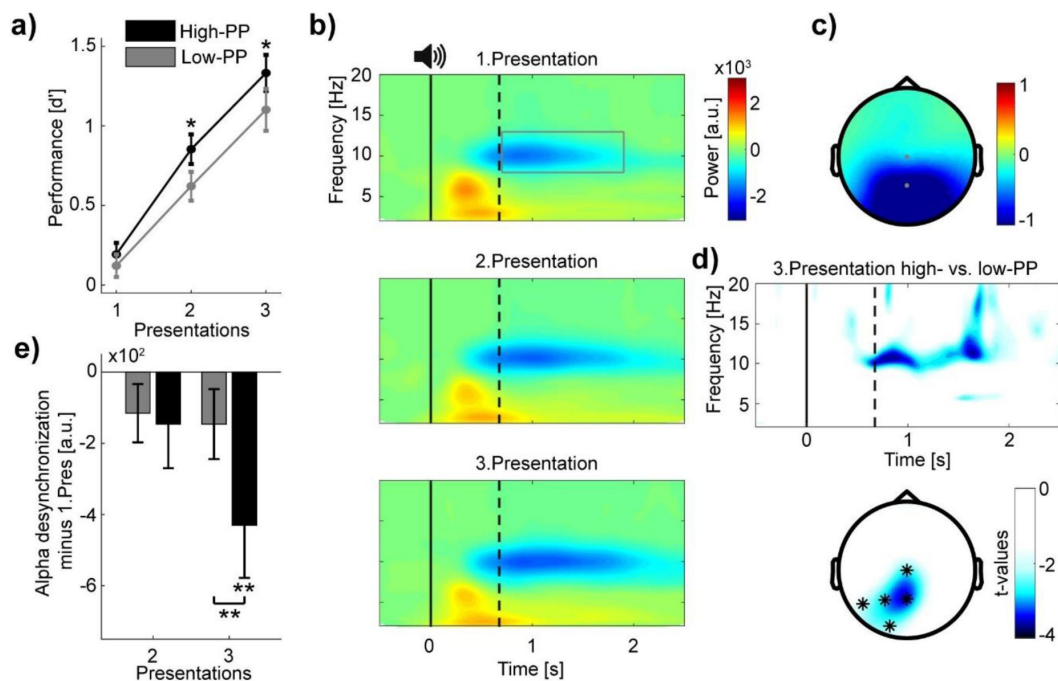


Fig. 2.

Distinct levels of encoding between high- and low-PP words.

a) Learning curves showing encoding performance over presentations of high- (black) and low-PP (gray) words. Note significant greater performance of high- in comparison to low-PP over the second and third presentations. b) Grand average time-frequency plots time-locked to word presentations. The gray rectangle within the top panel borders time and frequency range of interest (0.7 to 1.9s; 8-13Hz). Three different panels from top to down regarding to the three presentations. Solid and dotted lines within plots representing stimulus onset and averaged offset, respectively. Note, increases of oscillatory desynchronization in the alpha range (8-13Hz) over the three presentations. c) Topographic map shows power values averaged over the time window and frequency range of interest. d) (Top) Time-frequency representation of *t*-values (merged over Pz and P3 electrodes) shows significant greater changed desynchronization in alpha oscillations of high- in contrast to low-PP during the third presentation. Below, topographic map indicates significant cluster of electrodes of comparison between PP conditions of the third presentation (0.7 to 1.9s; 8-13Hz). e) The bar chart shows mean changes across subjects in alpha power (merged over Pz and Cz electrodes) of the second and third presentation by subtracting the first presentation in high- (black) vs. low-PP (gray). Statistical analyses revealed significant higher desynchronization of high- compared to low-PP and a significant decrease in alpha power under 0 of high-PP at the third presentation. Error bars reflect standard errors of the Mean (SEM); * $p < 0.05$, ** $p < 0.01$.

of encoding depth^{35,36}. To summarize, as a manipulation check based on encoding analyses, we confirmed that the conditions of high- and low-PP correspond to distinct levels in learning difficulty.

TMR affects memory consolidation of the easy to learn words

To examine whether TMR during sleep impacts memory consolidation of discrimination learning with respect to learning difficulty, we calculated the overnight changes by subtracting the pre- from the post-sleep memory performance based on d' -values of the reactivated sequences (cued) and non-reactivated sequences (uncued). In group 1, the high-PP sequence was cued, and the low-PP sequence was uncued. In group 2, the low-PP sequence was cued, and the high PP sequence was uncued (see **Fig. S5** for pre- and post-sleep memory test data of the different conditions). Conducting t -tests against 0 revealed a significant increase in the TMR/cueing condition of group 1 (high-PP cued: 0.24 ± 0.1 , $t(10) = 2.44$, $p = 0.035$, Cohen's $d = 0.74$), while the memory performance remained unchanged in all the other conditions (low-PP cued: -0.09 ± 0.08 , $t(10) = -1.16$, $p = 0.27$; high-PP uncued: 0 ± 0.08 , $t(10) = -0.04$, $p = 0.97$; low-PP uncued: 0.03 ± 0.1 , $t(10) = 0.31$, $p = 0.76$; **Fig. 3a**).

An additional two-way mixed design ANOVA on the same values with the factor cueing (cued vs. uncued) as a within-subject factor and group as a between-subject factor revealed trends of significance ($p < 0.1$) for the interaction (cueing $\times$ group: $F(1,20) = 3.47$, $p = 0.08$) and the main effect of group ($F(1,20) = 3.28$, $p = 0.09$). The main effect of cueing was not significant ($F(1,20) = 0.58$, $p = 0.46$). Post-hoc pairwise comparisons revealed a significant difference between the conditions of high-PP-cued and low-PP-cued ($t(20) = 2.63$, $p = 0.02$, Cohen's $d = 1.12$), a trend of significance between high-PP-cued and high-PP-uncued ($t(20) = 1.91$, $p = 0.07$) and no significant difference between high-PP-cued and low-PP-uncued ($t(10) = 1.55$, $p = 0.15$). In additional control analyses, pre-sleep memory performance and vigilance shortly before the post-sleep memory task did not significantly (p -values ≥ 0.08) differ between both cueing groups (see **Table S1**).

We examined next as an exploratory analysis whether TMR conditions influence biases in decision-making. Therefore, we analyzed changes between the pre- and post-sleep memory tasks of c -criterion values as a measurement of risk avoidance/seeking in decision making³⁴. Here, we found as well a significant overnight increase for cued sequences in group 1 (high-PP cued: 0.2 ± 0.07 , $t(10) = 3.04$, $p = 0.01$, Cohen's $d = 0.92$), indicating an increased bias towards risk avoidance by TMR (**Fig. 3b**), while all the other conditions remained unchanged (high-PP uncued: 0.09 ± 0.06 , $t(10) = 1.4$, $p = 0.19$; low-PP cued: 0.08 ± 0.05 , $t(10) = 1.63$, $p = 0.13$; low-PP uncued: 0.03 ± 0.05 , $t(10) = 0.56$, $p = 0.59$). An ANOVA on c -criterion changes showed no significant effects (interaction cueing $\times$ group: $F(1,20) = 2.66$, $p = 0.12$; main effect cueing $F(1,20) = 2.08$, $p = 0.17$; main effect group $F(1,20) = 0.38$, $p = 0.55$).

Taken together, these results suggest that the effectiveness of TMR depends on the level of difficulty in word learning, while auditory cueing during sleep increases the memory performance of the easy to learn words.

Increased spindle power nested during slow wave up-states in TMR of the easy to learn words

After analyzing TMR's effectiveness on behavior, we investigated the corresponding neural activities by EEG. By visual inspection of the signals, auditory word presentations during NREM sleep led to broad high-amplitude oscillations, called slow waves (SW; 0.5-3Hz), whereas sleep spindle activity (9-16Hz) with various amplitude occurred preferentially nested during the SW's up-state phase (see for example traces **Fig. 4a**).

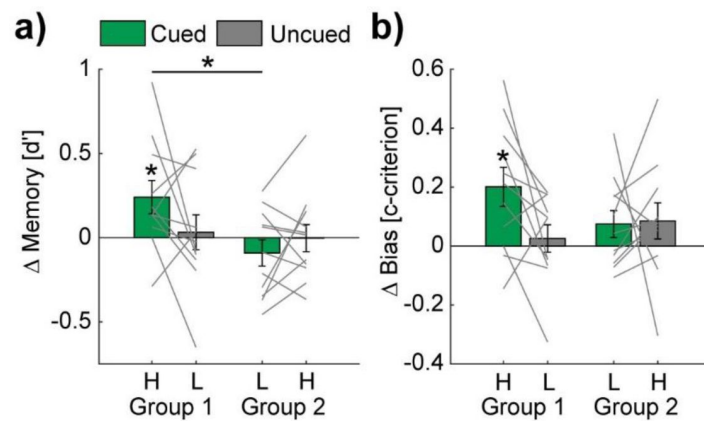


Fig. 3.

TMR affects the memory performance of the easy to learn words.

Bar charts show mean overnight changes of d' (a) and c-criterion (b) values of high (H) - and low (L) - PP and cued (green) vs. uncued (gray) conditions. Note, statistical analyses revealed significant overnight increases only in the high-PP cued condition. Gray lines represent individual data points. Error bars reflect SEM; * $p < 0.05$.

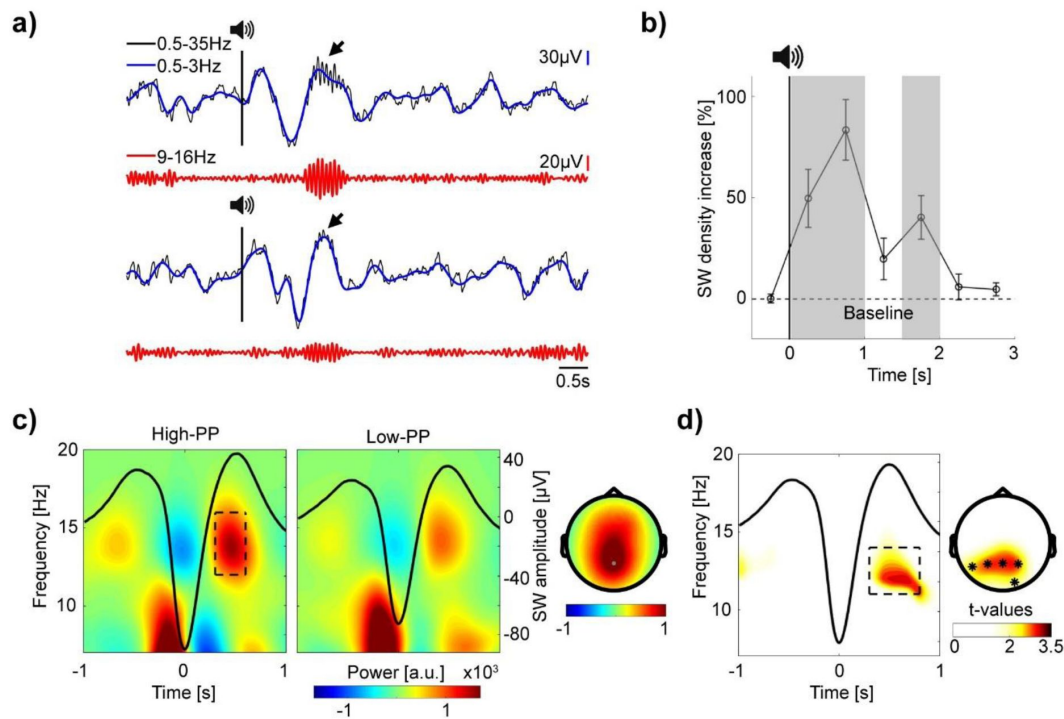


Fig. 4.

Increased spindle power during SW up-states in TMR of the easy to learn words.

a) Top and bottom panel, two example EEG traces of auditory cueing during sleep (~ 2 until 6s to stimulus onset). Top rows, in blue, signal filtered in the SW range (0.5-3Hz) superimposed upon the broadband (0.5-35Hz) signal in black. Vertical black lines with speaker symbols on top mark onsets of auditory presentations. Black arrows point to spindle activity during SW up-states. Bottom rows, in red, the same signal, but filtered in the spindle range (9-16Hz). Note, elevated SW following cueing presentations with various spindle band activity nested during SW up-states. b) Grand average baseline corrected curve of increased SW density after TMR in percentage. Shaded gray areas mark time windows (0 to 0.5s, 0.5 to 1s and 1.5 to 2s) of significant increased SW density. c) Grand average time-frequency plots time-locked to the troughs of SW with averaged signals plotted as black lines. Two different panels (left and right) according to high- vs. low-PP cueing conditions. The rectangle within the left panel borders time (0.3 until 0.6s to SW troughs) and frequency range of up-state fast spindle band activity (12-16Hz). Corresponding topographic map at right shows elevated fast spindle power over mid-parietal electrodes. d) Time-frequency representation of t -values time-locked to SW shows significant greater spindle band power during SW up-states for high- vs. low-PP (merged over Pz, P3 and P4 electrodes). Right, topographic map of t -values shows corresponding significant cluster of electrodes (0.3 to 0.8s; 11-14Hz), $*p < 0.05$.

To statistically analyze whether TMR increases the SW density, we conducted a detection algorithm. To control for individual differences, we included 30% of SW with the highest amplitude per subject (see methods) in subsequent analyses. Testing changes of SW density in time windows of 0.5s, revealed significant increases after stimulus onset in comparison to baseline (**Fig. 4b**; baseline period of -1.5 to 0s before stimulus onset; t -tests against 0; 0 to 0.5s: $49.73 \pm 14.32\%$, $t(21) = 3.47$, $p = 0.002$, Cohen's $d = 0.74$; 0.5 to 1s: $83.44 \pm 14.97\%$, $t(21) = 5.57$, $p < 0.001$, Cohen's $d = 1.19$; 1.5 to 2s: $40.33 \pm 10.67\%$, $t(21) = 3.78$, $p = 0.001$, Cohen's $d = 0.84$; see **Table S6** for all statistical comparisons of time-bins from -0.5 to 3s and **Fig. S7** for SW distribution from -0.5 to 6s time-locked to stimulus onset). Additional comparisons between the TMR conditions of high- and low-PP showed no significant differences in SW density, SW amplitude, number of TMR presentations and other sleep parameters, but a trend of significance for REM sleep parameters (see **Table S2**). In sum, these results indicate an increase of SW in response to auditory word stimulation independent of difficulty in word learning.

Next, we analyzed whether variations in spindle-band oscillations nested during the SW up-states after auditory TMR – as a neural signature of memory reactivation – differ between the conditions of high- and low-PP. To extract the spindle power during SW up-states, we conducted time-frequency analyses time-locked to the troughs of the detected SW in the time-window from 0 to 6s after auditory presentations. **Fig. 4c** shows the grand average analyses per condition, whereas the fast spindle band activity (12-16Hz) nested during SW up-states displayed a topography with maximum power over the central-parietal electrodes. Comparison between PP conditions revealed a significant greater spindle-band power prominent in the frequency range of 11-14Hz during SW up-states (from 0.3 to 0.8s after the SW trough) of high- in contrast to low-PP (averaged t -value over the 5 cluster electrodes = 2.66, $p = 0.01$, Cohen's $d = 1.13$; **Fig. 4d**), whereas post-hoc comparison including the individual SW amplitude as a covariate still revealed the significance of this cluster ($t(20) = 2.38$, $p = 0.03$, Cohen's $d = 1.02$).

By conducting an additional analyses based on detection of fast sleep spindles (12-16Hz; see methods), we confirmed that fast sleep spindles during the SW up-states (from 0.3 to 0.8s after the SW trough) occurred with significantly higher amplitude after the cueing presentation of high- compared to low-PP words, whereas parameters of sleep spindle density and the amount sleep spindles coupled to the SW up-state did not differ between the cueing conditions (see **Fig. S8** and **Table S9**).

These results suggest that successful TMR of the easy to learn words is associated with increased spindle activity nested during SW up-states.

Whereas we found a significant group difference in spindle power nested during SW up-states, conducting further whole sample ($n = 22$) correlation analyses between the individual spindle power values of the significant cluster and the overnight changes of behavior measurements revealed no significant correlations ($\Delta d'$: $r = 0.16$, $p = 0.48$; Δc -criterion: $r = 0.19$, $p = 0.40$).

Discussion

In this study, we asked whether learning difficulty of artificial words affects TMR's effectiveness and its neural activities. After confirming easier encoding of artificial words with high- compared to low phonotactic familiarity, we found that auditory TMR during sleep improved the memory performance of the easy to learn words, whereas cueing of the difficult words had no effect. Correspondingly, at the neural level, we observed that cueing presentations of the easy words induced significant greater spindle-band oscillations nested during SW up-states in comparison to the difficult words. To our knowledge, we present here the first study to address the critical role of learning difficulty by experimental manipulations of phonotactical word properties on TMR's effectiveness and its neural signatures.

Related to our finding of TMR's effectiveness of the easy words, previous research in object association learning showed beneficial influences of encoding depth²⁹ and prior knowledge²⁸ on TMR. Language studies without using TMR revealed that prior knowledge affected sleep dependent memory integration of new words in children³⁷ and in adults^{18,20}. To explain these critical influences on consolidation during sleep, the model of the complementary systems approach of word learning (CLS)⁴ proposes that new word memory temporarily encoded in the hippocampus transfer during sleep into cortical structures of lexical and phonological memory for long-term embedding, due to co-activation of both memory systems. Thus, word memory reactivation and consolidation during sleep depend on the hippocampal encoding depth and the cortical prior knowledge in phonotactics. Regarding our results, we suggest facilitated systems co-activation/consolidation of the easy to learn words by auditory cueing during sleep as a manipulation of both factors - via enhanced initial encoding and higher similarity to pre-existing phonotactical knowledge - compared to the difficult to learn words.

Contrary to our results, other studies found better memorization of words with low phonological¹⁶ and semantical¹⁷ similarity to pre-existing knowledge after a retention period of sleep compared to words with high similarity. To explain this discrepancy, we could speculate that a very high degree of similarity led to a strong memory formation during wake-encoding, which prevented an additional effect of consolidation during sleep. Indeed, Creery et al.²⁹ showed that learned object-location associations only benefited from TMR during sleep when not almost perfectly recalled during prior wake. Here, a moderate encoding depth might benefit most from subsequent sleep consolidation compared to insufficient or almost perfect encoding. Alternatively, the contrary results could be related to characteristics of the learning paradigms (L1- vs. L2-like novel words with associations to objects¹⁶; semantically related vs. unrelated word pairs¹⁷) distinct to our design.

By conducting an additional exploratory analysis, we observed a significant change of the decision bias in the cueing condition of the easy to learn words and no overnight changes in the other conditions. Here, subjects became more biased in their decision making to avoid losses of money points. This observation suggests that TMR not only captures specific memory traces of the words and their reward associations, but also decision-making tendencies. In a previous study, Ai and colleagues³⁸ revealed altered reward-relevant decision preferences by TMR. To what extent concepts of decision-making such as risk avoidance/seeking could be reactivated, transformed, and consolidated by TMR is largely unexplored and might be interesting for future research for example in neuroeconomics.

Correspondent to our behavior results of TMR's effectiveness, we found significantly higher spindle-band oscillations nested during SW up-states after cueing presentations of the easy in comparison to the difficult words. Accordingly, the active system consolidation hypothesis proposes slow wave - sleep spindle coupling activities as an underlying mechanism of systems consolidation^{6,12,39}. In addition to our result of the significant group difference, we failed to find significant correlations between SW nested spindle power values and overnight changes in behavior measurements, whereas previous studies reported associations of SW and spindle activities during sleep with the integration of new memories in pre-existing knowledge networks^{40,41}. Strikingly, recent studies by implementing machine learning for EEG decoding revealed that classification accuracy of distinct memory categories peaked above chance level synchronized with SW-spindle coupling events^{26,42}. Here, our results support the assumption of SW-spindle activities as an underlying mechanism of memory reactivation during sleep by providing additional evidence to link successful TMR on behavior to increased spindle activity during SW up-states.

In validation of our novel paradigm, we confirmed easier encoding and better memorization of artificial words with high- in comparison to low-PP, in line with previous studies^{1,2}. These results suggest that participants used their prior knowledge of frequently occurring word-sound

patterns to facilitate learning. Regarding the framework of schema theory³, processing of incoming words with high-PP might co-activate consistent schemata of prior word sound knowledge and thus may lead to facilitated memory integration and stabilization during encoding in comparison to processing of more inconsistent words with low-PP.

Our EEG analyses of encoding revealed desynchronization of alpha oscillations as a function of learning improvements. Additionally, superior encoding of the high-PP words corresponded to greater alpha desynchronization in comparison to the low-PP words. Consequently, we assume that the suppression of alpha oscillations over parietal and occipital electrodes represent a neural correlate of mnemonic functioning. Accordingly, previous studies linked desynchronization of alpha oscillations to memory processing^{35,43} and memory replay during wake⁴⁴. A study of intracranial electrophysiology in epilepsy patients⁴⁵ showed that neocortical alpha/beta desynchronization precede and predict fast hippocampal gamma activity during learning. The authors propose that these coupled activities represent hippocampal-cortical information transfer important for memory formation. Here, our findings let suggest cortical alpha desynchronization as neural marker of wake mnemonic processing according to distinct levels of difficulty in word learning. Additional studies linked alpha desynchronization to cognitive effort and cognitive load^{46,47}. So, one could assume to observe higher alpha desynchronization in the more difficult to learn condition of low-PP compared to high-PP. On the other hand numerous studies investigating oscillatory correlates of learning and memory showed that alpha desynchronization is associated with memory across different tasks, modalities and experimental phases of encoding and retrieval^{35,36,44,45,48,49}. Strikingly, Griffith and colleagues⁴⁹ revealed by simultaneous EEG-fMRI recordings a negative correlation between the occurrence of patterns of stimulus-specific information detected by fMRI and cortical alpha/beta suppression. Here, the authors suggested that a decrease of alpha/beta oscillations might represent the neuronal mechanism of unmasking the task-critical signal by simultaneous suppression of task-irrelevant neuronal activities to promote information processing. Following this interpretation, we assume that over the course of learning elevated memory processing of the easier to learn stimuli is associated with enhanced information processing and thus accompanied by higher cortical alpha desynchronization in comparison of the more difficult to learn stimuli.

Our study has limitations due to a small sample size and between-subject comparisons. The criteria of data analyses were not pre-registered and the *p*-values of our behavior analyses were not corrected for multiple comparisons. Further, potential influences of phonotactic word properties and the prior encoding depth on TMR cannot be interpreted independently because the conditions of phonotactic word properties impacted already the encoding depth before sleep. To disentangle these potential factors, future studies could include a higher number of encoding trials for the low-PP words to equalize the encoding depth between conditions prior to TMR, as successfully shown in a recent study²⁰. Additionally, a within-subject design with a larger sample size would provide a more robust control of interindividual differences in sleep and cognition. To control for potential confounders despite the influence of difficulty in word learning on TMR, we compared parameters of sleep, the pre-sleep memory performance and the vigilance shortly before the post-sleep memory test, revealing no significant group differences (see **Table S1** and **S2**). Nevertheless, we cannot rule out that other individual trait factors differed between the groups, such as the individual susceptibility to TMR. To rule out these alternative explanations based on individual factors, we suggest for future research to replicate our study by conducting a within-subject design with cueing of subsets of previously learned low- and high-PP words providing all conditions within the same individuals as shown in other TMR studies^{25,26}. Further, we used artificial words based on American English in combination with German speaking participants, whereas language differences of pronunciation and phoneme structures might affect word perception and memory processing⁵⁰. On the other hand, both languages are considered to have the same language family⁵¹ and the phonological distance between English and German is quite short compared for example to Korean⁵². Thus, major common phonological characteristics across both languages are still preserved. In addition, our

behavior analyses revealed robust word discrimination learning and distinct memory performance according to different levels of phonotactic probabilities providing evidence of successful experimental manipulation. Based on our paradigm, we investigated categorization learning of artificial words according to their reward associations (rewarded vs. unrewarded) and did not studied aspects of generalization learning of artificial grammar rules⁵³–⁵⁵. This difference might limit the comparability between these previous language-related studies and our findings. However, the usage of artificial words with distinct phonotactical properties provided a successful way to manipulate learning difficulty and to investigate word properties on TMR, whereas our reward categorization learning paradigm had the advantage to increase the relevance of the word learnings due to incentives.

To conclude, our present study demonstrates that difficulty in word learning as a manipulation of phonotactic properties impacts the effectiveness of TMR. Here, TMR of the easy to learn words facilitates categorization learning while TMR of the difficult words had no effect. Auditory cueing of the easy words induced increased spindle activity during the SW up-states - as an assumed underlying activity of memory reactivation - compared to the difficult words. Thus, our findings suggest that future research and clinical applications to restore language capabilities should consider learning difficulty based on phonotactical word properties as a potential factor of successful TMR, whereas alpha desynchronization during initial learning - as a neural marker of encoding depth - may serve as a predictor for sleep-associated memory reactivation and consolidation.

Methods

Subjects

The study included 39 subjects (29 females) with an age range of 19 to 28 years ($M = 22.28$ years, $SD = 2.04$). Participants were recruited from the University of Fribourg community by E-Mail or through advertisements at the campus of the University. Before participation, subjects had to give written informed consent as approved by the Ethical Commission of the Department of Psychology of the University of Fribourg. All participants were German speakers, and no subject had a history of neurological or psychiatric illness. The participants were instructed to keep a normal sleep schedule, to get up in the morning before 8 a.m. and not to consume alcohol and caffeine on experimental days. For participation, subjects either received credit for an undergraduate class and/or monetary compensation.

Data were excluded from subjects who did not reach the minimal learning performance of $d' > 1.05$ during the pre-sleep memory test in at least one of the two PP conditions, whereas this threshold value corresponds to accuracy rates of 70% ($n = 5$). In addition, we excluded one subject who showed a negative d' in one PP condition of the pre-sleep memory test ($n = 1$).

Pre-Learning

Participants arrived at the sleep laboratory at 19:30h. Electrodes for standard polysomnography (32 EEG electrodes, EMG and ECG electrodes) were mounted with two EEG mastoid electrodes. One electrode under the right eye was attached to record eye movements.

Encoding task

The encoding task started around 21:00h. Subjects learned to discriminate rewarded and unrewarded artificial words by right- and left-hand button presses. As a two alternative forced-choice task, we assigned left- and right-hand button presses to the rewarded and the unrewarded word category, counterbalanced across subjects. We instructed the participants to respond to each word by left- or right-hand button presses, whereas one button means the word is rewarded (gain

of money points) and the other button means the word is unrewarded (avoid the loss of money points). During each trial, an initial fixation cross was displayed between 1.7 and 2.3s. Subsequently, concurrent with the onset of a blank screen the sound of an artificial word was played for 0.7 to 1s. After a waiting period (blank screen 2s), a question mark appeared to signal the onset of the response time window (maximal duration 4s). Following the key press response, a feedback screen with the money points of the trial and the current task score appeared for 2s. The learning task contained 480 trials in randomized order according to 160 artificial words with three presentations each.

Pre-Sleep Memory task

After finishing the encoding task, following a 10min break, subjects performed the pre-sleep memory task. The memory task had the same trial structure as the encoding task without the last feedback screen. Participants received again money points for correct responses. However, the received amount was only shown at the end of the task. The memory task contains 160 trials according to the 160 artificial words.

Auditory target memory reactivation during sleep

Following an additional impedance check and re-adjustment of the EEG electrodes, subjects went to bed in a noise and electric shielded cabin of the sleep laboratory. All night sleep periods started with light off between 23:00 and 24:00h. Based on online monitoring of N2 and N3 sleep, artificial words were presented aurally via loudspeakers (sound pressure level 55dB) with a randomized inter-stimulus interval of 8 ± 2 s. One group of subjects were exposed to artificial words with low phonotactic probability during sleep, whereas we presented to the other group the high phonotactic probability words (see [Table S1](#) [↗](#) for number of reactivations). We interrupted word presentations when we observed online an arousal or patterns of REM sleep.

Post-Sleep Memory task

After sleep and re-adjustment of the EEG, subjects performed the post-sleep memory task (see above the description of the pre-sleep memory task).

Experimental tasks

All experimental tasks, including sleep reactivation, were conducted by using the E-Prime software (Psychology Software Tools, Sharpsburg, USA). By presentation of practice trials at the beginning, the subject's understanding of the task was approved. Across all tasks, stimuli were presented in randomized order.

Stimuli

To create artificial words, we subdivided the alphabet in two groups of consonants (C1: b, c, d, f, g, h, j, k, l, m; C2: n, p, q, r, s, t, v, w, x, z) and vowels (V1: a, e, i; V2: o, u, y). Four-letter-words were created by selecting letters from the vowel and consonant groups according to four different sequences (G1: C1, V1, V2, C2; G2: C1, V1, C2, V2; G3: V1, C1, C2, V2; G4: V1, C1, V2, C2). Artificial words were converted automatically from text to speech files (wav format) by using MATLAB functions (tts.m, audiowrite.m) with the setting of a female computer voice with American English pronunciation. From this pool of 900 stimuli for each rule, we selected 40 artificial words per rule category (4×40 words) according to the inclusion criteria: fluent and understandable pronunciation; no meaning; no names; aurally discriminable to other selected words (see [Table S3](#) [↗](#) and [S4](#) [↗](#) for lists of the used words).

Phonotactic probability values were computed for each word by using an online computation platform⁵⁶ [↗](#) (<https://calculator.ku.edu/phonotactic/about> [↗](#)) according to an American English lexicon. Between rule category comparison analyses revealed significant differences in

phonotactic probabilities between two rules in contrast to the other two rules (see **Fig. 1b** [↗](#)).

Psychomotor vigilance task

With respect to vigilance after sleep before the post-sleep memory task, a reaction time measurement was conducted by the Psychomotor Vigilance Task (PVT) [57](#) [↗](#). Each trial began with a fixation cross of a randomized duration between 2 and 10 seconds. Participants were instructed to press the spacebar button with the forefinger of the non-dominant hand as quickly as possible after numbers started to count in milliseconds on the screen. The reaction time was displayed for 1s after the key press. The PVT had a duration of 10 min (see for results **Table S2** [↗](#)).

Behavior analyses

As a measurement of discrimination learning and memory performance, we calculated d' -values ($d' = z(\text{hits}) - z(\text{false alarms})$) according to the signal detection theory [34](#) [57](#) [↗](#). Here, we considered correct trials with gaining of money points as hits and incorrect trials with a loss of money points as false alarms (see **Fig. 1d** [↗](#)). In addition, we assessed measurements of the response bias ($c = -0.5 * (z(\text{false alarms}) + z(\text{hits}))$). According to the convention [58](#) [↗](#), rates of 0 and 1 were replaced with $0.5/n$ and with $(n-0.5)/n$ respectively, whereas n corresponds to the number of trials ($n = 40$). Values of d' and response bias were calculated separately for the low- and high-PP condition.

EEG recordings

We made EEG recordings by using customized 32-Ag/AgCl electrodes at 10-10 locations caps (EASYCAP, Woerthsee-Ettersschlag, Germany) and 32 channel amplifiers (Brain Products, Gilching, Germany). Impedances were kept below 10 k Ω . The EEG was recorded with a sampling rate of 500Hz using Brain Vision Recorder software (Brain Products, Gilching, Germany). Signals were referenced to electrodes at the mastoids. The ocular activity was measured via one EOG channel mounted ~2 cm below the right eye. Muscle tone was monitored by EMG recordings made under the chin. The following EEG electrodes were used for subsequent wake and sleep analyses: Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T7, T8, P7, P8, Fz, Cz and Pz.

Sleep scoring

The sleep stages of NREM 1 to 3 (N1 to N3), wake, and REM sleep were scored offline and manually according to the criteria of the American Academy of Sleep Medicine (AASM) by visual inspection of the signals of the frontal, central, and occipital electrodes over 30s epochs [59](#) [↗](#). Based on offline scoring, we confirmed TMR exposure during N2 and N3 and no significant differences (p -values > 0.05) of sleep parameters between the cueing groups (see **Table S2** [↗](#)).

Preprocessing Wake EEG

Wake EEG were preprocessed using the Fieldtrip toolbox (<http://fieldtriptoolbox.org> [↗](#); Donders Institute for Brain, Cognition and Behaviour, Radboud University, Netherlands) [60](#) [↗](#). First, raw data signals were re-referenced to averaged mastoid electrodes and filtered (high- and low-pass, 0.5 and 45Hz). For each trial, data were segmented into epochs (-2 to 4s) time-locked to stimulus onset. After demean and detrend processing steps, noisy trials were identified by visual inspection and discarded from further analyses. We then conducted an independent component analysis (ICA) to identify and reject ICA components impacted by eye blinks and eye movements.

Preprocessing Sleep EEG

EEG signals during sleep were re-referenced to averaged mastoid electrodes and filtered (high- and low-pass, 0.5 and 35Hz). To analyze EEG activities of TMR, data were segmented into epochs (-2 to 8s) around the onset of stimuli presentations. After demean and detrend processing, trials with signals distorted by noise and movement artefacts were identified by visual inspection and removed from further analyses.

Time-Frequency Analysis Wake data

We conducted the time-frequency analyses by using Morlet wavelet analyses implemented in the Fieldtrip toolbox. The number of wavelet cycles was adjusted to 7 cycles. We extracted oscillatory power in the frequency range of 1 to 45Hz with frequency steps of 0.2Hz and time steps of 10ms. All the power values were baseline corrected and transformed to absolute changed values by subtracting the corresponding averaged values of the baseline interval of -1 to -0.1s before stimulus onset.

Slow wave-spindle power analyses

To reduce influence of potential confounders on TMR like low prior learning performance and number of reactivations, we included only subjects with a pre-sleep memory performance $d' > 0.75$ in both PP conditions and who did receive a minimal number of 160 reactivations (each word was represented at least two times during sleep). Thus, we excluded 5 and 6 subjects respectively. The final sample for sleep analyses consisted of $n = 22$ with 11 subjects in each cueing condition.

To detect slow waves, we first localized all negative and positive peaks of the 0.5 to 3Hz band-pass filtered signal during the auditory cueing trials in the time windows of 0 to 6s according to stimulus onsets. In addition, time distance between the prior and posterior positive peaks should be in the range of 0.3 to 2s. For the following analyses, the 30% slow waves with the highest amplitude (from negative peak to posterior positive peak) were included for each subject.

We then extracted the corresponding spectral power (1-30Hz) in a time window around this negative slow wave peak ($\pm 2s$) using Morlet wavelet (7 cycles) analysis implemented in the Fieldtrip toolbox, with a time window and step size of 10ms and frequency steps of 0.2Hz. Extracted power data over frequency and time was averaged of the slow waves and baseline corrected by subtracting the averaged values of the baseline interval of -2 to -1.5s prior to the negative slow wave peak. Finally, the transformed absolute changed values were plotted for the frequency and time range of interest and different conditions (see [Fig. 4c](#) [↗](#)).

Slow wave density analyses

To analyze the number of slow waves around memory reactivations during sleep, we divided the time course of reactivation trials in equally sized bins with a duration of 0.5s from -1.5 to 3s according to stimulus onset. Based on negative peaks of detected slow waves, we counted the occurring of slow waves for each bin over trials. Subsequently, we computed density values for each bin by division of the number of trials. Finally, we calculated the relative increase in slow wave density in percent by division of the averaged baseline values (bins: -1.5 to 0s; see [Fig. 4b](#) [↗](#)). In an additional histogram, we show the grand average slow wave density distribution in number per trial without baseline correction and with a bin size of 0.5s from -0.5 to 6s time-locked to stimulus onset (see [Fig. S7](#) [↗](#)).

Sleep spindle analyses

We detected fast sleep spindles by band-pass filtering (12-16Hz) the signal of the Pz electrode during the auditory cueing trials in the time windows of -2 to 8s according to stimulus onsets. The amplitude threshold was calculated individually for each subject as 1.25 standard deviations (SDs) from the mean. The beginning and end times of the sleep spindles were then defined as the points at which the amplitude fell below 0.75 SDs before and after the detected sleep spindle. Only sleep spindles with a duration of 0.5-3 s were included in subsequent analyses.

To compare the sleep spindle densities between the different cueing conditions of high- and low-PP, we computed the grand average sleep spindle density distribution in number per trial with a bin size of 0.5s from -0.5 to 6s time-locked to stimulus onset in each condition (see [Fig. S8a](#) [↗](#) and

Table S9 [↗](#)).

Based on the detected slow waves and sleep spindles, we defined coupling events when the positive amplitude peak of a detected sleep spindle was occurring during the slow wave up-state phase in a time window of 0.3 to 0.8s according to the trough of a slow wave.

We computed the averaged amplitude size of each detected sleep spindle by calculating the mean of the absolute amplitude values of all negative and positive peaks within a detected sleep spindle (see **Fig. S8b** [↗](#)).

Statistical Analysis

To test for differences between conditions of wake behavioral data, repeated-measure ANOVAs and paired sample *t*-tests were used. To analyze overnight changes of sleep behavioral data within TMR conditions, we conducted at first dependent sample *t*-tests against 0 of Δ -values (post-sleep test minus pre-sleep test) of d' and *c*-criterion (see **Fig. 3** [↗](#)). Two-way mixed design ANOVAs were computed to compare Δ -values between TMR conditions. After confirming at least a trend of significance ($p < 0.1$) for the interaction effect, we conducted post-hoc pairwise comparisons by independent and dependent sample *t*-tests. For all behavior statistical analyses, the *p*-value was set at $p < 0.05$ for two-tailed testing. A *p*-value < 0.1 and > 0.05 was reported as a trend of significance.

All statistical analyses were processed by using MATLAB 2018a (MathWorks, Natick, USA) and SPSS (IBM Corp., Version 25).

Considering the problem of multiple testing in statistics, we applied cluster based permutation tests implemented in the Fieldtrip toolbox⁶⁰ [↗](#) to test for potential differences of EEG time-frequency data. Here, dependent sample *t*-tests and an independent sample test (for between subject group comparison of sleep data) were conducted. We specified the number of permutations by 1000 and the minimal number of channels by two. The cluster alpha level was set at $p = 0.05$ for two-tailed testing. Plots with significant clusters of electrodes are shown in **Fig. 2d** [↗](#) and **4d** [↗](#). After confirming the existence of a significant cluster, we conducted an additional post-hoc repeated-measure ANOVA with averaged values of the identified time and frequency range of interest and merged over the Pz and Cz electrodes (see **Fig. 2e** [↗](#)). To control for SW amplitude size as a potential confounder in spindle activity differences during SW up-states, we extracted and averaged power values of the significant cluster (0.3 to 0.8s; 11-14Hz; Pz, P3, P4, O2, P7) per subject and conducted a post-hoc independent sample *t*-test based on residuals corrected by individual SW amplitude size. By using the same extracted power values (0.3 to 0.8s; 11-14Hz; Pz, P3, P4, O2, P7) per subject, we performed whole sample ($n = 22$) Pearson correlation analyses between these power values and the overnight changes of behavior measurements of the cued condition ($\Delta d'$ and Δc -criterion).

Supplementary data

	High-PP cued ($n = 11$)	Low-PP cued ($n = 11$)	t	P
Female (#)	10	7	-	-
Age (years)	21.91 ± 0.33	22.00 ± 0.56	-0.14	0.89
Pre-sleep memory test				
High-PP (d')	2.11 ± 0.16	2.03 ± 0.26	0.27	0.79
Low-PP (d')	1.58 ± 0.16	1.72 ± 0.23	-0.53	0.60
High-PP (c-criterion)	0.30 ± 0.09	0.57 ± 0.11	-1.86	0.08
Low-PP (c-criterion)	0.34 ± 0.09	0.60 ± 0.14	-1.60	0.13
Psychomotor vigilance task before post-sleep memory test				
Reaction time (ms)	319.97 ± 8.99	333.63 ± 10.19	-1.01	0.33

Data are means $\pm$ SEM. PP, phonotactic probability. P -values of statistical comparisons between groups by using unpaired t -tests. Note, no significant group differences, but a trend of significance for pre-sleep memory test of c-criterion values of the high-PP condition.

Supplementary table S1

Gender, age and performance rates

	High-PP cued (<i>n</i> = 11)	Low-PP cued (<i>n</i> = 11)	<i>t</i>	<i>P</i>
Duration (min)				
N1	91.55 ± 20.02	65.64 ± 18.77	0.94	0.36
N2	181.64 ± 11.61	200.55 ± 14.5	-1.02	0.32
N3	119.96 ± 15.11	99.46 ± 12.11	1.06	0.30
REM	57.91 ± 13.30	88.86 ± 9.87	-1.87	0.08
WASO	16.96 ± 10.18	5.55 ± 1.19	1.11	0.29
Duration (%)				
N1	19.44 ± 4.18	13.98 ± 3.89	0.96	0.35
N2	38.85 ± 2.56	43.77 ± 3.30	-1.18	0.25
N3	25.54 ± 3.16	21.55 ± 2.52	0.99	0.34
REM	12.40 ± 2.82	19.46 ± 2.20	-1.98	0.06
WASO	3.71 ± 2.27	1.23 ± 0.27	0.28	0.29
Number of reactivations				
N2	61.00 ± 22.22	93.18 ± 24.62	-0.97	0.34
N3	266.18 ± 53.68	231.73 ± 34.43	0.54	0.60
Slow wave detection				
SW density (#/3s)	1.87 ± 0.09	1.99 ± 0.13	-0.76	0.46
SW amplitude (μV)	134.33 ± 14.28	107.78 ± 9.96	1.53	0.14

Data are means ± SEM. N1, N2: Non-REM sleep stages N1, N2 and N3; REM, rapid-eye movement sleep; WASO, wake after sleep onset; SW, slow wave; PP, phonotactic probability. *P*-values of statistical comparisons between groups by using unpaired *t*-tests. Note, no significant group differences, but a trend of significance for REM sleep parameters.

Supplementary table S2

Sleep and reactivation parameter

<i>G1 (High-PP)</i>		<i>Phonotactic Probabilities</i>		<i>G2 (High-PP)</i>		<i>Phonotactic Probabilities</i>	
<i>Number</i>	<i>Word</i>	<i>Phoneme</i>	<i>Biphone</i>	<i>Word</i>	<i>Phoneme</i>	<i>Biphone</i>	
1	baos	0.180	0.005	baqy	0.126	0.004	
2	beon	0.146	0.003	basy	0.205	0.006	
3	biox	0.181	0.003	bavo	0.156	0.005	
4	biuq	0.098	0.002	beto	0.167	0.006	
5	cayr	0.114	0.000	bevo	0.125	0.004	
6	ceot	0.145	0.002	bisu	0.173	0.004	
7	ceov	0.070	0.001	bitu	0.160	0.005	
8	ceox	0.135	0.000	bixy	0.121	0.003	
9	cion	0.105	0.002	catu	0.145	0.004	
10	cioz	0.070	0.001	cipu	0.088	0.002	
11	daon	0.178	0.004	devo	0.126	0.004	
12	daot	0.220	0.004	disy	0.177	0.005	
13	deop	0.136	0.002	fasu	0.197	0.005	
14	deus	0.146	0.002	fato	0.194	0.006	
15	diov	0.117	0.004	faxo	0.152	0.002	
16	faop	0.162	0.002	gazu	0.118	0.003	
17	faoq	0.126	0.002	gexo	0.101	0.002	
18	faor	0.169	0.007	giny	0.168	0.006	
19	faot	0.215	0.004	haqy	0.114	0.004	
20	faox	0.206	0.002	haso	0.200	0.007	
21	faoz	0.138	0.002	heso	0.168	0.005	
22	gaor	0.148	0.007	hesu	0.159	0.004	
23	geon	0.121	0.003	hevo	0.113	0.003	
24	giur	0.115	0.000	jaqo	0.081	0.000	
25	gius	0.123	0.001	jiry	0.124	0.000	
26	giut	0.162	0.002	kapy	0.204	0.021	
27	haor	0.161	0.008	kexu	0.158	0.002	
28	heoq	0.087	0.001	kipo	0.183	0.004	
29	heox	0.167	0.001	kipu	0.173	0.002	
30	hion	0.137	0.003	kiqu	0.146	0.001	
31	jaoq	0.079	0.000	kizo	0.166	0.002	
32	jios	0.101	0.002	leqo	0.084	0.002	
33	kaon	0.219	0.018	lexu	0.099	0.002	
34	kaot	0.261	0.018	liwo	0.092	0.003	
35	kaov	0.187	0.017	manu	0.225	0.019	
36	kiuz	0.151	0.001	mawo	0.144	0.006	
37	leon	0.129	0.003	meqy	0.101	0.003	
38	leoq	0.082	0.002	mino	0.206	0.006	
39	liux	0.160	0.003	mipy	0.140	0.005	
40	miux	0.184	0.002	mivy	0.127	0.004	

Supplementary table S3

List of the high-PP words and their phoneme and biphone probabilities

Supplementary table S4

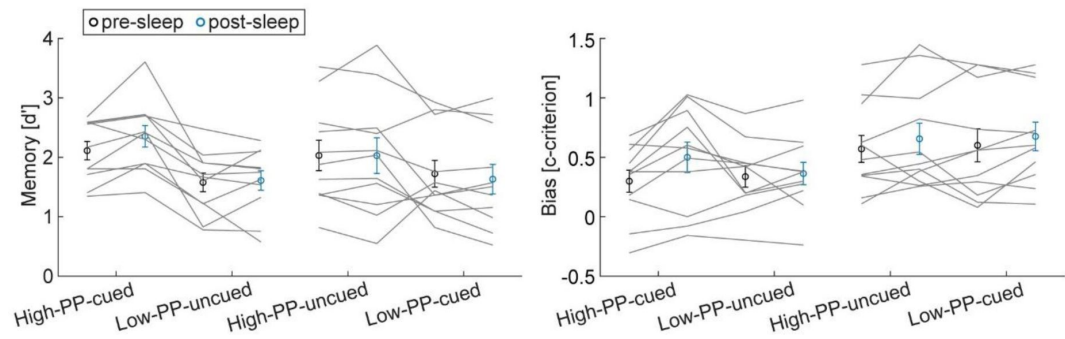
List of the low-PP words and their phoneme and biphone probabilities

Number	G3 (Low-PP)	Phonotactic Probabilities		G4 (Low-PP)	Phonotactic Probabilities	
	Word	Phoneme	Biphone	Word	Phoneme	Biphone
1	abpu	0.069	0.002	afut	0.121	0.002
2	abwo	0.047	0.002	afux	0.112	0.000
3	adro	0.120	0.003	agyv	0.035	0.000
4	adwo	0.047	0.000	ahon	0.078	0.001
5	adzo	0.062	0.001	ahun	0.074	0.001
6	afwo	0.043	0.000	ahuz	0.039	0.001
7	afzo	0.058	0.000	ajot	0.120	0.002
8	agno	0.135	0.002	alor	0.118	0.008
9	agvy	0.055	0.000	ebon	0.078	0.002
10	agzo	0.059	0.002	ebox	0.111	0.001
11	ahvy	0.050	0.000	ecyn	0.070	0.000
12	akvu	0.074	0.001	efon	0.074	0.002
13	alpo	0.115	0.002	ehos	0.073	0.001
14	alzu	0.088	0.000	ehyq	0.007	0.000
15	ampy	0.083	0.009	ehyt	0.096	0.000
16	ebry	0.105	0.001	ehyv	0.022	0.000
17	ebvy	0.050	0.001	ejyz	0.019	0.000
18	ecpy	0.072	0.002	ekoz	0.062	0.002
19	ecto	0.107	0.003	ekyr	0.077	0.001
20	edty	0.092	0.000	ekyz	0.046	0.001
21	edxu	0.048	0.001	elon	0.114	0.005
22	edxy	0.051	0.001	elyp	0.088	0.000
23	efny	0.119	0.003	elys	0.102	0.000
24	efqy	0.023	0.000	emos	0.092	0.002
25	efsy	0.102	0.000	emyn	0.073	0.001
26	efwy	0.028	0.000	ibon	0.077	0.002
27	egnu	0.117	0.001	icon	0.085	0.001
28	egro	0.109	0.004	icuc	0.035	0.000
29	egxy	0.048	0.000	idon	0.077	0.002
30	ekpo	0.089	0.002	idox	0.110	0.000
31	ibry	0.104	0.000	idoz	0.043	0.001
32	ibvu	0.047	0.000	idur	0.069	0.000
33	idno	0.129	0.001	ihos	0.072	0.001
34	idqo	0.033	0.000	ijut	0.108	0.002
35	igpy	0.060	0.002	ijyv	0.021	0.000
36	igru	0.098	0.002	iluq	0.063	0.002
37	igty	0.089	0.000	ilyt	0.140	0.000
38	igvo	0.053	0.001	ilyx	0.131	0.000
39	ikvu	0.066	0.001	imup	0.074	0.002
40	imno	0.140	0.001	imus	0.087	0.001

Supplementary Fig. S5.

Pre- and post-sleep memory test data.

Pre (black) and post (blue) sleep mean group memory test data of d' (left panel) and c-criterion (right panel) divided into the conditions of high- and low-PP and cued and uncued. Gray lines represent individual data points. Error bars reflect SEM.



Supplementary table S6

Statistics of SW density time-bin analyses from -0.5 to 3s

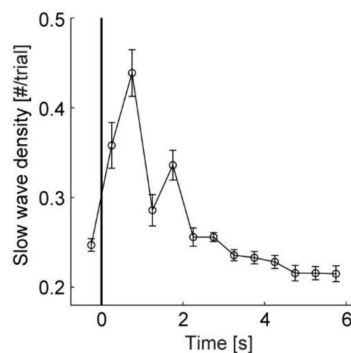
Time-bins (s)	SW density (%)	<i>t</i>	<i>P</i>
-0.5 to 0s	0.21 ± 2.10	0.10	0.92
0 to 0.5s	49.73 ± 14.32	3.47	0.002
0.5 to 1s	83.44 ± 14.97	5.57	<0.001
1 to 1.5s	19.77 ± 10.26	1.93	0.07
1.5 to 2s	40.33 ± 10.67	3.78	0.001
2 to 2.5s	5.98 ± 6.37	0.94	0.36
2.5 to 3s	4.81 ± 3.08	1.56	0.13

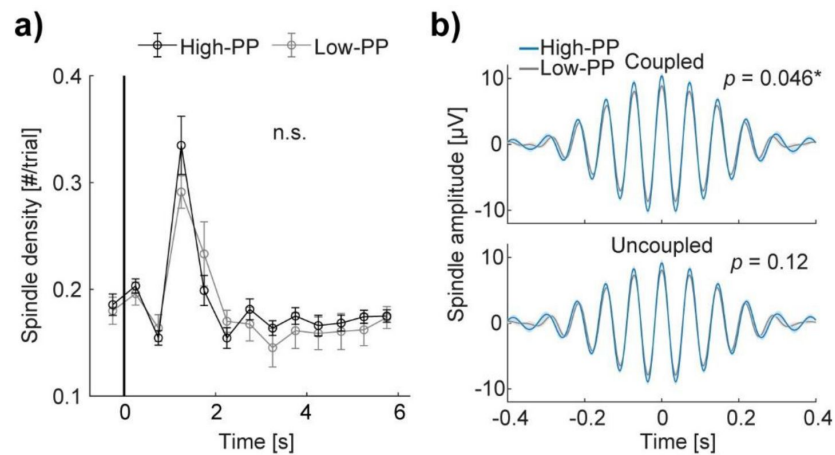
Data are means ± SEM. *P*-values (uncorrected for multiple comparisons) of statistical comparisons between groups by using paired *t*-tests against 0.

Supplementary Fig. S7.

Slow wave density distribution.

Grand average curve of slow wave density time-locked to stimulus onset in number per trial with a bin size of 0.5s from -0.5 to 6s. Note, slow wave density peaks during the time window of 0 to 2s and lower density values from 3 to 6s after stimulus onset.





Supplementary Fig. S8.

Detected sleep spindle analyses.

a) Grand average curves of sleep spindle density time-locked to stimulus onset in number per trial with a bin size of 0.5s from -0.5 to 6s of the cueing condition of high (black) - and low (gray) - PP, respectively. Note, sleep spindle density peaks during the time window of 1 to 1.5s after stimulus onset with no significant (n.s., $p > 0.05$) difference between conditions. b) Grand average signal filtered (12-16Hz) of the Pz electrode time-locked to the sleep spindle amplitude peak of the high (blue) - and low (gray) - PP cueing conditions; shaded areas represent SEM. Two different panels according to sleep spindles coupled (top) and uncoupled (bottom) with detected slow waves during the time window of 0 to 6s after stimulus onset. Note, significant higher amplitude sleep spindles coupled with slow waves in the cueing condition of high- compared to low-PP and no significant difference between the cueing conditions of the uncoupled sleep spindles. Error bars reflect SEM; $*p < 0.05$.

After conducting a sleep spindle detection (frequency range of 12-16Hz, see methods for details), we compared the sleep spindle density between the TMR conditions of high- and low-PP showing no significant difference (see **Fig. S8a** [↗](#) and **Table S9** [↗](#)). Next, we subdivided the detected sleep spindles into coupled and uncoupled sleep spindles with the previously detected slow waves (SW; analyses of **Fig. 4** [↗](#)). Sleep spindles were defined as coupled when their amplitude peak occurred during the SW up-state phase (0.3 to 0.8s time-locked to the SW troughs). A two-way mixed design ANOVA on the amplitude size of the sleep spindles with the cueing group as a between-subject factor (high-PP-cued vs. low-PP-cued) and SW-coupling as a within-subject factor (coupled vs. uncoupled) showed a significant interaction effect (cueing group $\times$ SW-coupling: $F(1,20) = 4.51$, $p = 0.046$, $\eta^2 = 0.18$), a significant main effect of SW-coupling ($F(1,20) = 85.02$, $p < 0.001$, $\eta^2 = 0.81$), and a trend of significance of the main effect of the cueing group ($F(1,20) = 3.54$, $p = 0.08$). Post-hoc unpaired t -tests revealed a significant higher amplitude size of the coupled sleep spindles of the cueing group of high-compared to low-PP ($t(20) = 2.13$, $p = 0.046$, Cohen's $d = 0.91$; **Fig. S8b** [↗](#)) and no significant group difference of the uncoupled sleep spindles ($t(20) = 1.62$, $p = 0.12$). An additional comparison of the amount of coupled sleep spindles between the cueing groups revealed no significant difference (see **Table S9** [↗](#)).

Here, we found that detected sleep spindles coupled to the SW up-state phase occurred with higher amplitude after TMR presentations of the high-PP words in comparison to the low-PP words, whereas the sleep spindle density and the amount of sleep spindles coupled to the SW up-state phase did not differ between the cueing conditions.

	High-PP cued ($n = 11$)	Low-PP cued ($n = 11$)	t	P
Detected sleep spindles				
General density (#/trial)	2.25 ± 0.03	2.18 ± 0.11	0.60	0.56
SW-coupled (%)	38.10 ± 1.19	37.51 ± 2.32	0.22	0.83

Data are means $\pm$ SEM. PP, phonotactic probability; SW, slow wave. General density: sleep spindle density during the time window of 0 to 6s after stimulus onset in number per trial. SW-coupled: sleep spindles coupled to SW up-states divided by the total number of detected sleep spindles during the time window of interest (0 to 6s after stimulus onset) in percent. P -values of statistical comparisons between groups by using unpaired t -tests.

Supplementary table S9

Density parameters of detected sleep spindles

Author contributions

Design of the study: A.-L.K. and B.R.; data acquisition: A.-L.K.; statistical analyses and interpretation of the data: A.-L.K. and B.R.; writing the manuscript: A.-L.K. and B.R.

Acknowledgements

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Conflict of interest

The authors declare that they have no conflict of interest.

Data availability statement

Data is available on request.

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Reviewer #1 (Public Review):

Summary:

The authors aim to consider the effects of phonotactics on the effectiveness of memory reactivation during sleep. They have created artificial words that are either typical or atypical and showed that reactivation improves memory for the latter but not the former.

Strengths:

This is an interesting design and a creative way of manipulating memory strength and typicality. In addition, the spectral analysis on both the wakefulness data and the sleep data is well done. The article is clearly written and provides a relevant and comprehensive of the literature and of how the results contribute to it.

Weaknesses:

- (1) Unlike most research involving artificial language or language in general, the task engaged in this manuscript did not require (or test) learning of meaning or translation. Instead, the artificial words were arbitrarily categorised and memory was tested for that categorisation. This somewhat limits the interpretation of the results as they pertain to language science, and qualifies comparisons with other language-related sleep studies that the manuscript builds on.
- (2) Participants had to determine whether words are linked with reward or omission of punishment (if correctly categorised). Therefore, the task isn't a mere item categorisation task (group A/B), but also involves the complicated effects of reward (e.g., reward/loss asymmetries as predicted by prospect theory). This is not, in itself, a flaw, but there isn't a clear hypothesis as to the effects of reward on categorisation, and therefore no real justification for this design. This aspect of the task may add unneeded complexity (at best) or some reward-related contamination of the results (at worst).
- (3) The study starts off with a sample size of $N=39$ but excludes 17 participants for some crucial analyses. This is a high number, and exclusion criteria were not pre-registered. Having said that, some criteria seem very reasonable (e.g., excluding participants who were not fully exposed to words during sleep).
- (4) Relatedly, the final N is low for a between-subjects study ($N=11$ per group). This is adequately mentioned as a limitation, but since it does qualify the results, it seemed important to mention it here.
- (5) The linguistic statistics used for establishing the artificial words are all based on American English, and are therefore in misalignment with the spoken language of the participants (which was German). This is a limitation of the study. The experimenters did not check whether participants were fluent in English. In all fairness, the behavioural effects presented in Figure 2A are convincing, providing a valuable manipulation test.

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Reviewer #2 (Public Review):**Summary:**

The work by Klaassen & Rasch investigates the influence of word learning difficulty on sleep-associated consolidation and reactivation. They elicited reactivation during sleep by applying targeted memory reactivation (TMR) and manipulated word learning difficulty by creating words more similar (easy) or more dissimilar (difficult) to our language.

In one group of participants, they applied TMR of easy words and in another group of participants, they applied TMR of difficult words (between-subjects design). They showed that TMR leads to higher memory benefits in the easy compared to the difficult word group. On a neural level, they showed an increase in spindle power (in the up-state of an evoked response) when easy words were presented during sleep.

Strengths:

The authors investigate a research question relevant to the field, that is, which experiences are actually consolidated during sleep. To address this question, they developed an innovative task and manipulated difficulty in an elegant way.

Overall, the paper is clearly structured, results and methods are described in an understandable way. The analyses approach is solid.

Weaknesses:**(1) Sample size**

For a between-subjects design, the sample size is too small ($N = 22$). The main finding (also found in the title "Difficulty in artificial word learning impacts targeted memory reactivation") is based on an independent samples t-test with 11 participants/group. The authors explicitly mention the small sample size and the between-subjects design as a limitation in their discussion. Nevertheless, making meaningful inferences based on studies with such a small sample size is difficult.

(2) Choice of task

Even though the task itself is innovative, there would have been tasks better suited to address the research question. The main disadvantage the task and the operationalisation of memory performance (d') have is that single-trial performance cannot be calculated. Consequently, choosing individual items for TMR is not possible.

Additionally, TMR of low vs. high difficulty is conducted between subjects (and independently of pre-sleep memory performance) which is a consequence of the task design.

<https://doi.org/10.7554/eLife.90930.2.sa2>

Reviewer #3 (Public Review):**Summary:**

In this study, the authors investigated the effects of targeted memory reactivation (TMR) during sleep on memory retention for artificial words with varying levels of phonotactical similarity to real words. The authors report that the high phonotactic probability (PP) words showed a more pronounced EEG alpha decrease during encoding and were more easily learned than the low PP words. Following TMR during sleep, participants who had been cued with the high PP TMR, remembered those words better than 0, whilst no such difference was found in the other conditions. Accordingly, the authors report higher EEG spindle band power during slow-wave up-states for the high PP as compared to low PP TMR trials. Overall, the authors conclude that artificial words which are easier to learn benefit more from TMR than those which are difficult to learn.

Strengths:

(1) The authors have carefully designed the artificial stimuli to investigate the effectiveness of TMR on words that are easy to learn and difficult to learn due to their levels of similarity with prior word-sound knowledge. Their approach of varying the level of phonotactic probability enables them to have better control over phonotactical familiarity than in a natural language and are thus able to disentangle which properties of word learning contribute to TMR success.

(2) The use of EEG during wakeful encoding and sleep TMR sheds new light on the neural correlates of high PP vs low PP both during wakeful encoding and cue-induced retrieval during sleep.

Weaknesses:

(1) The present analyses are based on a small sample and comparisons between participants rather than within participants. Considering that the TMR benefits are based on changes in memory categorization between participants, it could be argued that the individuals in the high PP group were more susceptible to TMR than those in the low PP group for reasons other than the phonotactic probabilities of the stimuli (e.g., these individuals might be more attentive to sounds in the environment during sleep). While the authors acknowledge the small sample size and between-subjects comparison as a limitation, these results should be interpreted with caution.

Impact:

This work is likely to contribute to the subfield of sleep and memory, and their experimental methods could provide a useful resource for those which investigate memory processing of linguistic material.

<https://doi.org/10.7554/eLife.90930.2.sa1>

Author response:

The following is the authors' response to the original reviews.

*Public Reviews:**Reviewer #1 (Public Review):**Summary:*

The authors aim to consider the effects of phonotactics on the effectiveness of memory reactivation during sleep. They have created artificial words that are either typical or atypical and showed that reactivation improves memory for the latter but not the former.

*Comment 1:**Strengths:*

This is an interesting design and a creative way of manipulating memory strength and typicality. In addition, the spectral analysis on both the wakefulness data and the sleep data is well done. The article is clearly written and provides a relevant and comprehensive of the literature and of how the results contribute to it.

We thank the reviewer for his/her positive evaluation of our manuscript.

*Comment 2:**Weaknesses:*

(1) Unlike most research involving artificial language or language in general, the task engaged in this manuscript did not require (or test) learning of meaning or translation. Instead, the artificial words were arbitrarily categorised and memory was tested for that categorisation. This somewhat limits the interpretation of the results as they pertain to language science, and qualifies comparisons with other language-related sleep studies that the manuscript builds on.

We thank the reviewer for this comment. We agree that we did not test for meaning or translation but used a categorization task in which we trained subjects to discriminate artificial words according to their reward associations (rewarded vs. non-rewarded). Previous language studies (Batterink et al., 2014; Batterink and Paller, 2017; Reber, 1967) used artificial words to investigate implicit learning of hidden grammar rules. Here, the language researchers studied generalization of the previously learned grammar knowledge by testing subject's ability to categorize correctly a novel set of artificial words into rule-congruent versus rule-incongruent words. These differences to our study design might limit the comparability between the results of previous language studies of artificial grammar learning and our findings. We discussed now this aspect as a limitation of our novel paradigm.

We added the following sentences to the discussion on p.14, ll. 481-488:

Based on our paradigm, we investigated categorization learning of artificial words according to their reward associations (rewarded vs. unrewarded) and did not studied aspects of generalization learning of artificial grammar rules (Batterink et al., 2014; Batterink and Paller, 2017; Reber, 1967). This difference might limit the comparability between these previous language-related studies and our findings. However, the usage of artificial words with distinct phonotactical properties provided a successful way to manipulate learning difficulty and to investigate word properties on TMR, whereas our reward categorization learning paradigm had the advantage to increase the relevance of the word learnings due to incentives.

Comment 3:

(2) The details of the behavioural task are hard to understand as described in the manuscript. Specifically, I wasn't able to understand when words were to be responded to with the left or right button. What were the instructions? Were half of the words randomly paired with left and half with right and then half of each rewarded and half unrewarded? Or was the task to know if a word was rewarded or not and right/left responses reflected the participants' guesses as to the reward (yes/no)? Please explain this fully in the methods, but also briefly in the caption to Figure 1 (e.g., panel C) and in the Results section.

We thank the reviewer for this comment and added additional sentences into the document to provide additional explanations. We instructed the participants to respond to each word by left- and right-hand button presses, whereas one button means the word is rewarded and the other button means the word is unrewarded. The assignment of left- and right-hand button presses to their meanings (rewarded versus unrewarded) differed across subjects. In the beginning, they had to guess. Then over trial repetitions with feedback at the end of each trial, they learned to respond correctly according to the rewarded/unrewarded associations of the words.

We added the following sentences to the results section on p.5, ll. 161-168:

As a two alternative forced-choice task, we assigned left- and right-hand button presses to the rewarded and the unrewarded word category, counterbalanced across subjects. We instructed the participants to respond to each word by left- or right-hand button presses, whereas one button means the word is rewarded (gain of money points) and the other button means the word is unrewarded (avoid the loss of money points). In the beginning, they had to guess. By three presentations of each word in randomized order and by feedback at the end of each trial, they learned to respond correctly according to the rewarded/unrewarded associations of the words (Fig. 1c).

We added the following sentences to the caption of Figure 1 on p.6, ll. 188-194:

As a two alternative forced-choice task, responses of left- and right-hand button presses were assigned to the rewarded and the unrewarded word category, respectively. The participants were instructed to respond to each word by left- or right-hand button presses, whereas one button means the word is rewarded (gain of money points) and the other button means the word is unrewarded (avoid the loss of money points). d) Feedback matrix with the four answer types (hits: rewarded and correct; CR, correct rejections: unrewarded and correct; misses: rewarded and incorrect; FA, false alarms: unrewarded and incorrect) regarding to response and reward assignment of the word.

We added the following sentences to the methods on p.19, ll. 687-692:

As a two alternative forced-choice task, we assigned left- and right-hand button presses to the rewarded and the unrewarded word category, counterbalanced across subjects. We instructed the participants to respond to each word by left- or right-hand button presses, whereas one button means the word is rewarded (gain of money points) and the other button means the word is unrewarded (avoid the loss of money points).

Comment 4:

(3) Relatedly, it is unclear how reward or lack thereof would translate cleanly into a categorisation of hits/misses/correct rejections/false alarms, as explained in the text and shown in Figure 1D. If the item was of the non-rewarded class and the participant got it correct, they avoided loss. Why would that be considered a correct rejection, as the text suggests? It is no less of a hit than the rewarded-correct, it's just the trial was set up in a way that limits gains. This seems to mix together signal detection nomenclature (in which reward is uniform and there are two options, one of which is correct and one isn't) and loss-aversion types of studies (in which reward is different for two types of stimuli, but for each type you can have H/M/CR/FA separably). Again, it might all stem from me not understanding the task, but at the very least this required extended explanations. Once the authors address this, they should also update Fig 1D. This complexity makes the results relatively hard to interpret and the merit of the manuscript hard to access. Unless there are strong hypotheses about reward's impact on memory (which, as far as I can see, are not at the core of the paper), there should be no difference in the manner in which the currently labelled "hits" and "CR" are deemed - both are correct memories. Treating them differently may have implications on the d' , which is the main memory measure in the paper, and possibly on measures of decision bias that are used as well.

We thank the reviewer for this comment giving us the opportunity to clarify. As explained in the previous comment, for our two alternative forced-choice task, we instructed the participants to press one button when they were thinking the presented word is rewarded and the other button, when they were thinking the word is unrewarded. Based on this instruction, we applied the signal detection theory (SDT), because the subjects had the task to detect when reward was present or to reject when reward was absent. Therefore, we considered correct responses of words of the rewarded category as hits and words of the unrewarded category as correct rejections (see Table below). However, the reviewer is correct because in addition to false alarms, we punished here the incorrect responses by subtraction of money points to control for alternative task strategies of the participants instead of reward association learning of words. We agree that further explanation/argumentation to introduce our nomenclature is necessary.

Author response table 1.

	Respond: Reward Present	Respond: Reward Absent
Reward Present	Hit	Miss
Reward Absent	False Alarm	Correct Rejection

We adjusted the results section on p.5, ll. 169-177:

To obtain a measurement of discrimination memory with respect to the potential influence of the response bias, we applied the signal detection theory (Green and Swets, 1966). Because, we instructed the participants to respond to each word by left- or right-hand button presses and that one button means reward is present whereas the other button means reward is absent, we considered correct responses of words of the rewarded category as hits and words of the unrewarded category as correct rejections. Accordingly, we assigned the responses with regard to the reward associations of the words to the following four response types: hits

(rewarded, correct); correct rejections (unrewarded, correct); misses (rewarded, incorrect); and false alarms (unrewarded, incorrect). Dependent on responses, subjects received money points (Fig. 1d).

Comment 5:

(4) The study starts off with a sample size of N=39 but excludes 17 participants for some crucial analyses. This is a high number, and it's not entirely clear from the text whether exclusion criteria were pre-registered or decided upon before looking at the data. Having said that, some criteria seem very reasonable (e.g., excluding participants who were not fully exposed to words during sleep). It would still be helpful to see that the trend remains when including all participants who had sufficient exposure during sleep. Also, please carefully mention for each analysis what the N was.

Our study was not pre-registered. Including all the subjects independent of low prememory performance, but with respect to a decent number of reactivations (> 160 reactivations, every word at least 2 times), resulted in a new dataset with 15 and 13 participants of the high- and low-PP cueing condition, respectively. Here, statistical analyses revealed no significant overnight change anymore in memory performance in the high-PP cueing condition (Δ memory (d'): $t(14) = 1.67$, $p = 0.12$), whereas the increase of the bias in decision making towards risk avoidance still remained significant (Δ bias (c-criterion): $t(14) = 3.36$, $p = 0.005$).

We modified and added the following sentences to the discussion on p.13, ll. 456-458:

Our study has limitations due to a small sample size and between-subject comparisons. The criteria of data analyses were not pre-registered and the p-values of our behavior analyses were not corrected for multiple comparisons.

Comment 6:

(5) Relatedly, the final N is low for a between-subjects study (N=11 per group). This is adequately mentioned as a limitation, but since it does qualify the results, it seemed important to mention it in the public review.

We agree with the reviewer that the small sample size and the between subject comparisons represent major limitations of our study. Accordingly, we now discussed these limitations in more detail by adding alternative explanations and further suggestions for future research to overcome these limitations.

We added the following sentences to the discussion about the limitations on p.14, ll. 465-488:

To control for potential confounders despite the influence of difficulty in word learning on TMR, we compared parameters of sleep, the pre-sleep memory performance and the vigilance shortly before the post-sleep memory test, revealing no significant group differences (see Table S1 and S2). Nevertheless, we cannot rule out that other individual trait factors differed between the groups, such as the individual susceptibility to TMR. To rule out these alternative explanations based on individual factors, we suggest for future research to replicate our study by conducting a within-subject design with cueing of subsets of previously learned low- and high-PP words providing all conditions within the same individuals as shown in other TMR studies (Cairney et al., 2018; Schreiner and Rasch, 2015).

Comment 7:

(6) The linguistic statistics used for establishing the artificial words are all based on American English, and are therefore in misalignment with the spoken language of the participants (which was German). The authors should address this limitation and discuss

possible differences between the languages. Also, if the authors checked whether participants were fluent in English they should report these results and possibly consider them in their analyses. In all fairness, the behavioural effects presented in Figure 2A are convincing, providing a valuable manipulation test.

We thank the reviewer pointing to the misalignment between the German-speaking participants and the used artificial words based on American English. Further, we did not assess the English language capability of the participants to control it as a potential confounder, whereas comparative control analyses revealed no significant differences between the both cueing groups in pre-sleep memory performance (see Table S1).

We now discussed these comments as limitations on p.14, ll. 473-481:

Further, we used artificial words based on American English in combination with German speaking participants, whereas language differences of pronunciation and phoneme structures might affect word perception and memory processing (Bohn and Best, 2012). On the other hand, both languages are considered to have the same language family (Eberhard et al., 2019) and the phonological distance between English and German is quite short compared for example to Korean (Luef and Resnik, 2023). Thus, major common phonological characteristics across both languages are still preserved. In addition, our behavior analyses revealed robust word discrimination learning and distinct memory performance according to different levels of phonotactic probabilities providing evidence of successful experimental manipulation.

Comment 8:

(7) With regard to the higher probability of nested spindles for the high- vs low-PP cueing conditions, the authors should try and explore whether what the results show is a general increase for spindles altogether (as has been reported in the past to be correlated with TMR benefit and sleep more generally) or a specific increase in nested spindles (with no significant change in the absolute numbers of post-cue spindles). In both cases, the results would be interesting, but differentiating the two is necessary in order to make the claim that nesting is what increased rather than spindle density altogether, regardless of the SW phase.

We conducted additional analyses based on detected sleep spindles to provide additional data according to this question.

We added the following section to the supplementary data on pp. 31-32, ll. 1007-1045:

After conducting a sleep spindle detection (frequency range of 12-16Hz, see methods for details), we compared the sleep spindle density between the TMR conditions of high- and lowPP showing no significant difference (see Fig. S8a and Table S9). Next, we subdivided the detected sleep spindles into coupled and uncoupled sleep spindles with the previously detected slow waves (SW; analyses of Fig. 4). Sleep spindles were defined as coupled when their amplitude peak occurred during the SW up-state phase (0.3 to 0.8s time-locked to the SW troughs). A two-way mixed design ANOVA on the amplitude size of the sleep spindles with the cueing group as a between-subject factor (high-PP-cued vs. low-PP-cued) and SW-coupling as a within-subject factor (coupled vs. uncoupled) showed a significant interaction effect (cueing group $\times$ SW-coupling: $F(1,20) = 4.51$, $p = 0.046$, $\eta^2 = 0.18$), a significant main effect of SW-coupling ($F(1,20) = 85.02$, $p < 0.001$, $\eta^2 = 0.81$), and a trend of significance of the main effect of the cueing group ($F(1,20) = 3.54$, $p = 0.08$). Post-hoc unpaired t-tests revealed a significant higher amplitude size of the coupled sleep spindles of the cueing group of high- compared to low-PP ($t(20) = 2.13$, $p = 0.046$, Cohen's $d = 0.91$; Fig. S8b) and no significant group difference of the uncoupled sleep spindles ($t(20) = 1.62$, $p = 0.12$). An additional comparison of the amount

of coupled sleep spindles between the cueing groups revealed no significant difference (see Table S9).

Here, we found that detected sleep spindles coupled to the SW up-state phase occurred with higher amplitude after TMR presentations of the high-PP words in comparison to the low-PP words, whereas the sleep spindle density and the amount of sleep spindles coupled to the SW up-state phase did not differ between the cueing conditions.

We added the following sentences to the methods on pp. 22-23, ll. 822-839:

Sleep spindle analyses

We detected fast sleep spindles by band-pass filtering (12-16Hz) the signal of the Pz electrode during the auditory cueing trials in the time windows of -2 to 8s according to stimulus onsets. The amplitude threshold was calculated individually for each subject as 1.25 standard deviations (SDs) from the mean. The beginning and end times of the sleep spindles were then defined as the points at which the amplitude fell below 0.75 SDs before and after the detected sleep spindle. Only sleep spindles with a duration of 0.5-3 s were included in subsequent analyses.

To compare the sleep spindle densities between the different cueing conditions of high- and low-PP, we computed the grand average sleep spindle density distribution in number per trial with a bin size of 0.5s from -0.5 to 6s time-locked to stimulus onset in each condition (see Fig. S8a and Table S9).

Based on the detected slow waves and sleep spindles, we defined coupling events when the positive amplitude peak of a detected sleep spindle was occurring during the slow wave upstate phase in a time window of 0.3 to 0.8s according to the trough of a slow wave.

We computed the averaged amplitude size of each detected sleep spindle by calculating the mean of the absolute amplitude values of all negative and positive peaks within a detected sleep spindle (see Fig. S8b).

We added the following sentences to the results on p.10, ll. 338-343:

By conducting an additional analyses based on detection of fast sleep spindles (12-16Hz; see methods), we confirmed that fast sleep spindles during the SW up-states (from 0.3 to 0.8s after the SW trough) occurred with significantly higher amplitude after the cueing presentation of high- compared to low-PP words, whereas parameters of sleep spindle density and the amount sleep spindles coupled to the SW up-state did not differ between the cueing conditions (see Fig. S8 and Table S9).

Reviewer #2 (Public Review):

Summary:

The work by Klaassen & Rasch investigates the influence of word learning difficulty on sleep-associated consolidation and reactivation. They elicited reactivation during sleep by applying targeted memory reactivation (TMR) and manipulated word learning difficulty by creating words more similar (easy) or more dissimilar (difficult) to our language. In one group of participants, they applied TMR of easy words and in another group of participants, they applied TMR of difficult words (between-subjects design). They showed that TMR leads to higher memory benefits in the easy compared to the difficult word group. On a neural level, they showed an increase in spindle power (in the up-state of an evoked response) when easy words were presented during sleep.

Comment 9:

Strengths:

The authors investigate a research question relevant to the field, that is, which experiences are actually consolidated during sleep. To address this question, they developed an innovative task and manipulated difficulty in an elegant way.

Overall, the paper is clearly structured, and results and methods are described in an understandable way. The analysis approach is solid.

We thank the reviewer for his/her positive evaluation of our manuscript.

Weaknesses:

Comment 10:

(1) Sample size

For a between-subjects design, the sample size is too small ($N = 22$). The main finding (also found in the title "Difficulty in artificial word learning impacts targeted memory reactivation") is based on an independent samples t-test with 11 participants/group.

The authors explicitly mention the small sample size and the between-subjects design as a limitation in their discussion. Nevertheless, making meaningful inferences based on studies with such a small sample size is difficult, if not impossible.

We agree with the reviewer that the small sample size and the between subject comparisons represent major limitations of our study. Accordingly, we now discussed these limitations in more detail by adding alternative explanations and further suggestions for future research to overcome these limitations.

We added the following sentences to the discussion about the limitations on p.14, ll. 465-473:

To control for potential confounders despite the influence of difficulty in word learning on TMR, we compared parameters of sleep, the pre-sleep memory performance and the vigilance shortly before the post-sleep memory test, revealing no significant group differences (see Table

S1 and S2). Nevertheless, we cannot rule out that other individual trait factors differed between the groups, such as the individual susceptibility to TMR. To rule out these alternative explanations based on individual factors, we suggest for future research to replicate our study by conducting a within-subject design with cueing of subsets of previously learned low- and high-PP words providing all conditions within the same individuals as shown in other TMR studies (Cairney et al., 2018; Schreiner and Rasch, 2015).

Comment 11:

(2) Choice of task

though the task itself is innovative, there would have been tasks better suited to address the research question. The main disadvantage the task and the operationalisation of memory performance (d') have is that single-trial performance cannot be calculated. Consequently, choosing individual items for TMR is not possible.

Additionally, TMR of low vs. high difficulty is conducted between subjects (and independently of pre-sleep memory performance) which is a consequence of the task design.

The motivation for why this task has been used is missing in the paper.

We used a reward task combined with TMR because previous studies revealed beneficial effects of reward related information on sleep dependent memory consolidation and reactivation (Asfestani et al., 2020; Fischer and Born, 2009; Lansink et al., 2009; Sterpenich et al., 2021). In addition, we wanted to increase the motivation of the participants, as they could receive additional monetary compensation according to their learning and memory task performances. Furthermore, we designed the task, with the overall possibility to translate this task to operant conditioning in rats (see research proposal: <https://data.snf.ch/grants/grant/168602>). However, the task turned out to be too difficult to translate to rats, whereas we developed a different learning paradigm for the animal study (Klaassen et al., 2021) of this cross-species research project.

We added the following sentence to the introduction on p.4, ll. 134-137:

To consider the beneficial effect of reward related information on sleep dependent memory consolidation and reactivation (Asfestani et al., 2020; Fischer and Born, 2009; Lansink et al., 2009; Sterpenich et al., 2021), we trained healthy young participants to categorize these words into rewarded and unrewarded words to gain and to avoid losses of money points.

Reviewer #3 (Public Review):

Summary:

In this study, the authors investigated the effects of targeted memory reactivation (TMR) during sleep on memory retention for artificial words with varying levels of phonotactical similarity to real words. The authors report that the high phonotactic probability (PP) words showed a more pronounced EEG alpha decrease during encoding and were more easily learned than the low PP words. Following TMR during sleep, participants who had been cued with the high PP TMR, remembered those words better than 0, whilst no such difference was found in the other conditions. Accordingly, the authors report higher EEG spindle band power during slow-wave up-states for the high PP as compared to low PP TMR trials. Overall, the authors conclude that artificial words that are easier to learn, benefit more from TMR than those which are difficult to learn.

Comment 12 & 13:

Strengths:

(1) The authors have carefully designed the artificial stimuli to investigate the effectiveness of TMR on words that are easy to learn and difficult to learn due to their levels of similarity with prior wordsound knowledge. Their approach of varying the level of phonotactic probability enables them to have better control over phonotactical familiarity than in a natural language and are thus able to disentangle which properties of word learning contribute to TMR success.

(2) The use of EEG during wakeful encoding and sleep TMR sheds new light on the neural correlates of high PP vs. low PP both during wakeful encoding and cue-induced retrieval during sleep.

We thank the reviewer for his/her positive evaluation of our manuscript.

Weaknesses:

Comment 14:

(1) The present analyses are based on a small sample and comparisons between participants. Considering that the TMR benefits are based on changes in memory categorization between participants, it could be argued that the individuals in the high PP group were more susceptible to TMR than those in the low PP group for reasons other than the phonotactic probabilities of the stimuli (e.g., these individuals might be more attentive to sounds in the environment during sleep). While the authors acknowledge the small sample size and between-subjects comparison as a limitation, a discussion of an alternative interpretation of the data is missing.

We agree with the reviewer that the small sample size and the between subject comparisons represent major limitations of our study. We thank the reviewer for this helpful comment and now discussed these limitations in more detail by adding alternative explanations and further suggestions for future research to overcome these limitations.

We added the following sentences to the discussion on p.14, ll. 465-473:

To control for potential confounders despite the influence of difficulty in word learning on TMR, we compared parameters of sleep, the pre-sleep memory performance and the vigilance shortly before the post-sleep memory test, revealing no significant group differences (see Table S1 and S2). Nevertheless, we cannot rule out that other individual trait factors differed between the groups, such as the individual susceptibility to TMR. To rule out these alternative explanations based on individual factors, we suggest for future research to replicate our study by conducting a within-subject design with cueing of subsets of previously learned low- and high-PP words providing all conditions within the same individuals as shown in other TMR studies (Cairney et al., 2018; Schreiner and Rasch, 2015).

Comment 15:

(2) While the one-tailed comparison between the high PP condition and 0 is significant, the ANOVA comparing the four conditions (between subjects: cued/non-cued, within-subjects: high/low PP) does not show a significant effect. With a non-significant interaction, I would consider it statistically inappropriate to conduct post-hoc tests comparing the conditions against each other. Furthermore, it is unclear whether the p-values reported for the t-tests have been corrected for multiple comparisons. Thus, these findings should be interpreted with caution.

We thank the reviewer for this comment giving us the opportunity to correct our analyses and clarify with additional description. Indeed, we investigated at first overnight changes in behavior performance within the four conditions, conducting t-tests against 0 of Δ -values of d' and c-criterion. Whereas for all our statistical analyses the p-value was set at $p < 0.05$ for two-tailed testing, we did not corrected the p-value of our behavior analyses for multiple comparisons. To investigate subsequently differences between conditions, we conducted additional ANOVAs. We agree with the reviewer that without significant of results of the ANOVA, post-hoc analyses should not be conducted. Taken in account as well the recommendation of reviewer 1, we included now only post-hoc pairwise comparisons when the interaction effect of the ANOVA revealed at least a trend of significance ($p < 0.1$).

We removed the following post-hoc analyses from the results section on p.9, ll. 291-295:

Additional post-hoc pairwise comparisons revealed a significant difference between the highPP cued and low-PP uncued (high-PP cued vs. low-PP uncued: $t(10) = 2.43$, $p = 0.04$), and no difference to other conditions (high-PP cued vs.: high-PP uncued $t(20) = 1.28$, $p = 0.22$; lowPP cued $t(20) = 1.57$, $p = 0.13$).

Further, we mentioned the lack of correction for multiple comparisons as a limitation of our results in the discussion on p.13, ll. 456-458:

The criteria of data analyses were not pre-registered and the p-values of our behavior analyses were not corrected for multiple comparisons.

We added the following sentences to the methods p.23, ll. 842-849:

To analyze overnight changes of sleep behavioral data within TMR conditions, we conducted at first dependent sample t-tests against 0 of Δ -values (post-sleep test minus pre-sleep test) of d' and c-criterion (see Fig. 3). Two-way mixed design ANOVAs were computed to compare Δ values between TMR conditions. After confirming at least a trend of significance ($p < 0.1$) for the interaction effect, we conducted post-hoc pairwise comparisons by independent and dependent sample t-tests. For all behavior statistical analyses, the p-value was set at $p < 0.05$ for two-tailed testing. A p-value < 0.1 and > 0.05 was reported as a trend of significance.

Comment 16:

(3) With the assumption that the artificial words in the study have different levels of phonotactic similarity to prior word-sound knowledge, it was surprising to find that the phonotactic probabilities were calculated based on an American English lexicon whilst the participants were German speakers. While it may be the case that the between-language lexicons overlap, it would be reassuring to see some evidence of this, as the level of phonotactic probability is a key manipulation in the study.

We thank the reviewer pointing to the misalignment between the German-speaking participants and the used artificial words based on American English. In line with this recommendation, we added a more outlined argumentation to the manuscript about the assumption of our study that major common phonetic characteristics across both languages are still preserved.

We now discussed these aspects on p.14, ll. 473-481:

Further, we used artificial words based on American English in combination with German speaking participants, whereas language differences of pronunciation and phoneme structures might affect word perception and memory processing (Bohn and Best, 2012). On the other hand, both languages are considered to have the same language family (Eberhard et al., 2019) and the phonological distance between English and German is quite short compared for example to Korean (Luef and Resnik, 2023). Thus, major common phonological characteristics across both languages are still preserved. In addition, our behavior analyses revealed robust word discrimination learning and distinct memory performance according to different levels of phonotactic probabilities providing evidence of successful experimental manipulation.

Comment 17:

(4) Another manipulation in the study is that participants learn whether the words are linked to a monetary reward or not, however, the rationale for this manipulation is unclear. For instance, it is unclear whether the authors expect the reward to interact with the TMR effects.

We used a reward task combined with TMR because previous studies revealed beneficial effects of reward related information on sleep dependent memory consolidation and reactivation (Asfestani et al., 2020; Fischer and Born, 2009; Lansink et al., 2009; Sterpenich et al., 2021). In addition, we wanted to increase the motivation of the participants, as they could receive additional monetary compensation according to their learning and memory task

performances. Furthermore, we designed the task, with the overall possibility to translate this task to operant conditioning in rats (see research proposal: <https://data.snf.ch/grants/grant/168602>). However, the task turned out to be too difficult to translate to rats, whereas we developed a different learning paradigm for the animal study (Klaassen et al., 2021) of this cross-species research project.

We added the following sentence to the introduction on p.4, ll. 134-137:

To consider the beneficial effect of reward related information on sleep dependent memory consolidation and reactivation (Asfestani et al., 2020; Fischer and Born, 2009; Lansink et al., 2009; Sterpenich et al., 2021), we trained healthy young participants to categorize these words into rewarded and unrewarded words to gain and to avoid losses of money points.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Comment 18:

(1) Please clearly define all linguistics terms - and most importantly the term "phonotactics" - at first use.

We thank the reviewer for this recommendation and we added the definition of phonotactics and further reduced the diversity of linguistic terms to improve readability.

We added the following sentences to the beginning of the introduction on p.3, ll. 72-76:

One critical characteristic of similarity to pre-existing knowledge in auditory word processing is its speech sound (phoneme) pattern. In phonology as the field of language specific phoneme structures, phonotactics determines the constraints of word phoneme composition of a specific language.

Comment 19:

(2) Some critical details about the methods should be included in the Results section to make it comprehensible. For example, the way the crucial differences between G1-4 words should be addressed in the Results, not only in Figure 1.

According to the recommendation, we added this information to the results section. We added the following sentences to the results section on p.4, ll. 145-154:

To study the impact of difficulty in word learning on TMR, we developed a novel learning paradigm. We formed four sets of artificial words (40 words per set; see Table S3 and S4) consisting of different sequences of two vowels and two consonants. Here, we subdivided the alphabet into two groups of consonants (C1: b, c, d, f, g, h, j, k, l, m; C2: n, p, q, r, s, t, v, w, x, z) and vowels (V1: a, e, i; V2: o, u, y). Four-letter-words were created by selecting letters from the vowel and consonant groups according to four different sequences (G1: C1, V1, V2, C2; G2: C1, V1, C2, V2; G3: V1, C1, C2, V2; G4: V1, C1, V2, C2; Fig. 1a; see methods for further details). Comparison analyses between the sets revealed significant differences in phonotactic probability (PP; Fig. 1b; unpaired t-tests: $G1 / G2 > G3 / G4$, $p < 0.005$, values of Cohen's $d > 0.71$).

Comment 20

(3) Was scoring done both online and then verified offline? If so, please note that.

We included now this information.

We adjusted the method section on p.21, ll. 765-769:

The sleep stages of NREM 1 to 3 (N1 to N3), wake, and REM sleep were scored offline and manually according to the criteria of the American Academy of Sleep Medicine (AASM) by visual inspection of the signals of the frontal, central, and occipital electrodes over 30s epochs (Iber et al., 2007). Based on offline scoring, we confirmed TMR exposure during N2 and N3 and no significant differences (p-values > 0.05) of sleep parameters between the cueing groups (see Table S2).

Comment 21:

(4) In Figure 2, please arrange the panel letters in an easier-to-read way (e.g., label upper right panel b with a different letter).

Now we rearranged the panel letters according to the recommendation.

We adjusted Figure 2 on p.8, ll. 242-258:

Comment 22

(5) In the first paragraph on TMR effects, please note which memory measure you are comparing (i.e., d').

We added this information according to the recommendation.

We adjusted the sentence of the results on p.8, ll. 260-263:

To examine whether TMR during sleep impacts memory consolidation of discrimination learning with respect to learning difficulty, we calculated the overnight changes by subtracting the pre- from the post-sleep memory performance based on d'-values of the reactivated sequences (cued) and non-reactivated sequences (uncued).

Comment 23:

(6) Please show the pre-sleep and post-sleep test scores for both word categories (not only the delta). It may be best to show this as another data point in Fig 2a, but it may be helpful to also see this split between cued and uncued.

We added the pre-sleep and post-sleep test scores with the individual data points as an additional figure.

We added the following figure to the supplementary data on p.28, ll. 936-940:

Comment 24:

(7) In the sentence "An additional two-way mixed design ANOVA on the same values with cueing as a between-subject factor (cued vs. uncued) ...", a more exact phrasing for the last parentheses would probably be "(high-PP-Cued vs Low-PP-Cued)". Both groups were cued.

We thank the reviewer pointing this out. According to the recommendation, we corrected the descriptions of the two-way mixed design ANOVAs. In addition, we detected a mistake of wrong assignments of the conditions to ANOVAs and corrected the reported values.

We adjusted the sentences and corrected the values on p.9, ll. 271-275 and ll. 289-291:

An additional two-way mixed design ANOVA on the same values with the factor cueing (cued vs. uncued) as a within-subject factor and group as a between-subject factor revealed trends of significance ($p < 0.1$) for the interaction (cueing $\times$ group: $F(1,20) = 3.47$, $p = 0.08$) and the main effect of group ($F(1,20) = 3.28$, $p = 0.09$). The main effect of cueing was not significant ($F(1,20) = 0.58$, $p = 0.46$).

An ANOVA on c-criterion changes showed no significant effects (interaction cueing $\times$ group: $F(1,20) = 2.66$, $p = 0.12$; main effect cueing $F(1,20) = 2.08$, $p = 0.17$; main effect group $F(1,20) = 0.38$, $p = 0.55$).

Comment 25:

(8) In the same ANOVA, please mention that there is a trend toward an interaction effect. If there wasn't one, the post-hoc comparison would be unwarranted. Please consider noting other $p < 0.1$ p-values as a trend as well, for consistency.

Regarding this recommendation, we included now only post-hoc pairwise comparisons after confirming at least a trend toward an interaction effect of these ANOVAs and reported consistently a p-value < 0.1 and > 0.05 as a trend of significance.

We added the following sentences to the methods p.23, ll. 844-849:

Two-way mixed design ANOVAs were computed to compare Δ -values between TMR conditions. After confirming at least a trend of significance ($p < 0.1$) for the interaction effect, we conducted post-hoc pairwise comparisons by independent and dependent sample t-tests. For all behavior statistical analyses, the p-value was set at $p < 0.05$ for two-tailed testing. A p-value < 0.1 and > 0.05 was reported as a trend of significance.

We removed the following post-hoc analyses from the results section on p.9, ll. 291-295:

Additional post-hoc pairwise comparisons revealed a significant difference between the highPP cued and low-PP uncued (high-PP cued vs. low-PP uncued: $t(10) = 2.43$, $p = 0.04$), and no difference to other conditions (high-PP cued vs.: high-PP uncued $t(20) = 1.28$, $p = 0.22$; lowPP cued $t(20) = 1.57$, $p = 0.13$).

Comment 26:

(9) Please consider adding an analysis correlating spindle power with memory benefit across participants. Even if it is non-significant, it is important to report given that some studies have found such a relationship.

According to this recommendation, we conducted an additional correlation analyses.

We added the following sentences to the manuscript into the results (pp. 10-11, ll. 346-349), the discussion (p.12, ll. 413-417), and the methods (p.23, ll. 864-867):

Whereas we found a significant group difference in spindle power nested during SW up-states, conducting further whole sample ($n = 22$) correlation analyses between the individual spindle power values of the significant cluster and the overnight changes of behavior measurements revealed no significant correlations ($\Delta d'$: $r = 0.16$, $p = 0.48$; Δ c-criterion: $r = 0.19$, $p = 0.40$).

In addition to our result of the significant group difference, we failed to find significant correlations between SW nested spindle power values and overnight changes in behavior measurements, whereas previous studies reported associations of SW and spindle activities

during sleep with the integration of new memories in pre-existing knowledge networks (Tamminen et al., 2013, 2010).

By using the same extracted power values (0.3 to 0.8s; 11-14Hz; Pz, P3, P4, O2, P7) per subject, we performed whole sample ($n = 22$) Pearson correlation analyses between these power values and the overnight changes of behavior measurements of the cued condition ($\Delta d'$ and Δc criterion).

Reviewer #2 (Recommendations For The Authors):

(1) Choice of task

Comment 27:

In general, I find your task well-designed and novel. In light of your research question, however, I wonder why you chose this task. When you outlined the research question in the introduction, I expected a task similar to Schreiner et al. (2015). For example, participants have to associate high PP words with each other and low PP words. The advantage here would be that you could test the benefits of TMR in a within-subjects design (for example, cueing half of the remembered high and half of the remembered low PP words).

Please see our previous response at comment 14.

Comment 28:

Why did you decide to introduce a reward manipulation?

Please see our previous response at comment 11.

Comment 29:

Why did you do the cueing on a category level (cueing all high PP or all low PP words instead of single word cueing or instead of cueing 20 reward high-PP, 20 unrewarded high-PP plus 20 reward low-PP and 20 unrewarded low-PP)? Both alternatives would have provided you the option to run your statistics within participants.

Please see our previous response at comment 14.

Comment 30:

(2) Between-subjects design and small sample size.

Why did you decide on a between-subjects design that severely reduces your power?

Why did you just collect 22 participants with such a design? Were there any reasons for this small sample size? Honestly, I think publishing a TMR study with healthy participants and such a small sample size (11 participants for some comparisons) is not advisable.

Please see our previous response at comment 14.

Comment 31:

(3) Encoding performance.

Is d' significantly above 0 in the first repetition round? I would assume that the distinction between rewarded and non-rewarded words is just possible after the first

| round of feedback.

Indeed, conducting t-tests against 0 revealed significantly increased d'-values in the first repetition round (2nd presentation) in both PP conditions (high-PP: 0.85 ± 0.09 , $t(32) = 9.17$, $p < 0.001$; low-PP: 0.62 ± 0.09 , $t(32) = 6.83$, $p < 0.001$).

Comment 32:

(4) Encoding response options

If you want to you could make it more explicit what exactly the response options are. I assume that one button means a word has a high reward and the other button means a word has a low reward. Making it explicit increases the understanding of the results section.

Please see our previous response at comment 3.

Comment 33:

(5) Alpha desynchronisation.

Relative change

Why did you subtract alpha power during the 1st presentation from alpha power during 2nd and 3rd presentation? You baseline-corrected already and individually included the 1st, 2nd, and 3rd repetition in your behavioural analysis.

Based on this analysis, we aimed to examine the relative change in alpha power between PP-conditions of memory-relevant word repetitions. Therefore, to extract memory relevant changes of EEG activities, the first word presentation of naive stimulus processing could serve as a more representative baseline condition covering the time-window of interest of 0.7 to 1.9 s after the stimulus onset compared to a baseline condition before stimulus onset (-1 to -0.1s).

To explain the rational of the analyses with the baseline condition more clearly, we added this information to the results section on p.7, ll. 222-226:

We obtained the changes in power values by subtracting the first from the second and third presentation for the high- and low-PP condition, respectively. Here, the first word presentation of naive stimulus processing served us with a more representative baseline condition covering the time-window of interest of 0.7 to 1.9 s after the stimulus onset to examine relevant changes of encoding.

Comment 34:

(6) Alpha desynchronisation as a neural correlate of encoding depth & difficulty?

"In addition to the behavior results, these EEG results indicate differences between PP conditions in desynchronization of alpha oscillations, as an assumed neural correlate of encoding depth. In addition to the behavior results, these EEG results indicate differences between PP conditions in desynchronization of alpha oscillations, as an assumed neural correlate of encoding depth."

Given that the low-PP words are more difficult to learn, I was expecting to see higher alpha desynchronisation in the low-PP relative to the high-PP words. Could you outline in a bit more detail how your findings fit into the literature (e.g., Simon Hanslmayr did a lot of work on this)?

I would also advise you to add citations e.g., after your sentence in the quote above ("as an assumed neural correlate of encoding depth").

We thank the reviewer for the recommendation giving us the opportunity to discuss in more detail how our results relate to previous findings.

We added additional sentences to the discussion on p.13, ll. 441-455:

Additional studies linked alpha desynchronization to cognitive effort and cognitive load (Proskovec et al., 2019; Zhu et al., 2021). So, one could assume to observe higher alpha desynchronization in the more difficult to learn condition of low-PP compared to high-PP. On the other hand numerous studies investigating oscillatory correlates of learning and memory showed that alpha desynchronization is associated with memory across different tasks, modalities and experimental phases of encoding and retrieval (Griffiths et al., 2016, 2021, 2019a, 2019b; Hanslmayr et al., 2009; Michelmann et al., 2016). Strikingly, Griffiths and colleagues (Griffiths et al., 2019a) revealed by simultaneous EEG-fMRI recordings a negative correlation between the occurrence of patterns of stimulus-specific information detected by fMRI and cortical alpha/beta suppression. Here, the authors suggested that a decrease of alpha/beta oscillations might represent the neuronal mechanism of unmasking the task-critical signal by simultaneous suppression of task-irrelevant neuronal activities to promote information processing. Following this interpretation, we assume that over the course of learning elevated memory processing of the easier to learn stimuli is associated with enhanced information processing and thus accompanied by higher cortical alpha desynchronization in comparison of the more difficult to learn stimuli.

In addition, we added the mentioned quote on p.7, ll. 239-240:

In addition to the behavior results, these EEG results indicate differences between PP conditions in desynchronization of alpha oscillations, as an assumed neural correlate of encoding depth (Griffiths et al., 2021; Hanslmayr et al., 2009).

Comment 35:

(7) Exclusion criterion.

Why did you use a $d' > 0.9$ as a criterion for data inclusion?

This criterion ensured that each included subject had at least in one PP-condition a $d' > 1.05$ of pre-sleep memory performance, which corresponds to a general accuracy rate of 70%.

Accordingly, we adjusted these sentences of the method section on p.19, ll. 677-680:

Data were excluded from subjects who did not reach the minimal learning performance of $d' > 1.05$ during the pre-sleep memory test in at least one of the two PP conditions, whereas this threshold value corresponds to accuracy rates of 70% ($n = 5$). In addition, we excluded one subject who showed a negative d' in one PP condition of the pre-sleep memory test ($n = 1$).

Comment 36:

(8) Coherence of wording.

When you talk about your dependent variable (d') you sometimes use sensitivity. I would stick to one term.

We replaced the word sensitivity with d' .

(9) Criterion

Comment 37:

Why do you refer to a change in criterion (Figure 3b, axis labels) as a change in memory? Do you think the criterion says something about memory?

We corrected the axis label of Figure 3b and deleted here the word memory.

Comment 38:

Additionally, why did you analyse the effect of TMR on the criterion? Do you expect the criterion to change due to sleep-dependent memory consolidation? This section would benefit from more explanation. Personally, I am very interested in your thoughts and your hypothesis (if you had one, if not that is also fine but then, make it explicit that it was an exploratory analysis).

By conducting exploratory analyses of overnight changes of the c-criterion measurements, we aimed to examine the bias of decision-making to provide comprehensive data according to the framework of the signal detection theory. Regarding the previous literature showing mainly beneficial effects of sleep on learning and memory, we focused with our hypothesis on d' and explored additionally the c-criterion.

Despite our task design with gains/hits of +10 money points and losses/FAs of -8 (instead of -10), the subjects showed already during the pre-sleep memory task significant biases towards loss avoidance in both PP conditions (t-tests against 0: high-PP: 0.44 ± 0.07 , $t(21) = 5.63$, $p < 0.001$; low-PP: 0.47 ± 0.09 , $t(21) = 5.51$, $p < 0.001$). As already reported in the preprint, we found an additional significant increase of c-criterion by TMR solely for the high-PP words (see Fig. 3b). Even by integrating subjects with poor pre-sleep memory performance (high-PP-cueing group: $n = 15$; low-PP-cueing group: $n = 13$), t-tests against 0 revealed a significant increase of the high-PP cueing condition ($t(14) = 3.36$, $p = 0.005$) and no significant overnight changes in the other conditions (high-PP uncued: $t(12) = 1.39$, $p = 0.19$; low-PP cued: $t(12) = 1.47$, $p = 0.17$; low-PP uncued: $t(14) = -0.20$, $p = 0.84$). These exploratory findings on c-criterion suggest potential applications of TMR to affect decision-making biases in combination with reward learning.

We revised the manuscript mentioning the exploratory character of the c-criterion analyses of the results on p.9, ll. 282-283 and of the discussion on p.12, ll. 400-402:

We examined next as an exploratory analysis whether TMR conditions influence biases in decision-making.

By conducting an additional exploratory analysis, we observed a significant change of the decision bias in the cueing condition of the easy to learn words and no overnight changes in the other conditions.

Comment 39:

(10) You detected SWs in the time range of 0-6 sec post sound stimulation. How was the distribution of all detected SW down-states in this time range? (You could plot a histogram for this.)

We illustrated now the detected SWs in the time range of 0 to 6 s after stimulus onset.

We added a histogram to the supplementary section on p.30, ll. 982-986:

Reviewer #3 (Recommendations For The Authors):

Comment 40:

(1) In line with the weakness outlined above, I would recommend including a discussion of how the between-subject comparison and small sample size could affect the results and provide alternative interpretations.

Please see our previous response at comment 14.

Comment 41:

(2) Regarding my point about statistical comparisons, I would recommend that the authors follow best practice guidelines for post-hoc tests and multiple comparisons. In Figures 3a and b, I would also recommend removing the stars indicating significance from the post-hoc tests (if this is what they reflect). Perhaps this link will be useful: <https://www.statology.org/anova-post-hoc-tests/>

Please see our previous response at comment 15.

Comment 42:

(3) Furthermore, to address any doubts about the possible phonotactic probability differences between languages, I would recommend that the authors show whether the languages overlap, the level of English fluency in the German-speaking participants, and/or another way of reassuring that this is unlikely to have affected the results.

Please see our previous response at comment 7.

Comment 43:

(4) In the introduction, I would recommend that the authors outline a clear rationale for the reward/no reward manipulation.

Please see our previous response at comment 11.

Comment 44:

(5) Figure 1c: Please include what response options participants had, e.g., 'rewarded/not rewarded'. This would make the type of categorization clearer to the reader.

Please see our previous response at comment 3.

Comment 45:

(6) It is unclear whether the additional ANOVA conducted on the time and frequency of the identified clusters included all channels or only the channels contributing to the cluster. Consider clarifying this in the relevant methods and results. Furthermore, I would recommend labelling this as a posthoc test as this analysis was guided by an initial peak at the data and the timings, frequencies, and channels of interest were not selected a-priori.

We thank the reviewer for this recommendation and labelled the additional repeatedmeasure ANOVA as a post-hoc test. Further, we mentioned the used channels (Pz and Cz) for this analyses.

We adjusted the results section on p.7, ll. 230-233 and the methods section on p.23, ll. 858-860:

A post-hoc repeated-measure ANOVA on alpha power changes (merged over Pz and Cz electrodes) with PP (high vs. low) and presentations (2 to 3) as within-subjects factors revealed a main effect of PP ($F(1,32) = 5.42$, $p = 0.03$, $\eta^2 = 0.15$), and a significant interaction ($F(1,32) = 7.38$, $p = 0.01$, $\eta^2 = 0.19$; Fig. 2e).

After confirming the existence of a significant cluster, we conducted an additional post-hoc repeated-measure ANOVA with averaged values of the identified time and frequency range of interest and merged over the Pz and Cz electrodes (see Fig. 2e).

Comment 46:

(7) Figure 3: To better illustrate within- vs. between-subjects comparisons and promote transparency, please add individual points and lines between the within-subjects conditions.

According to this recommendation, we changed Figure 3 to add the individual data points by lines.

We modified Figure 3 on p.9, ll. 299-303:

Comment 47:

(8) For the SW density time-bin analyses, please include statistics for all comparisons (i.e., through 0 s to 3 s) and say whether these were corrected for multiple comparisons.

According to this recommendation, we included now statistics for all comparisons.

We added table S6 table to the supplementary data on p.29, l.962:

Comment 48:

(9) Consider reporting effect sizes.

We thank the reviewer for this recommendation and we added now effect sizes of significant results.

Comment 49:

(10) For transparency and replicability, consider including a list of the four stimulus sets including their phoneme and biphone probabilities.

We included a list of the four stimulus sets with their phoneme and biphone probabilities

We added table S3 and table S4 to the supplementary data on pp. 26-27:

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